Leveraged Learning

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This work rests on three insights.

1. Learning is the incremental ability to answer questions correctly that follow a pattern.
2. The number of questions reviewed is not the right measure of a learner's progress. What counts is the information received from asking them. Not all questions are created equal: about some, the learner already knows a great deal a priori, and their answers count for less.
3. One can infer *more* than a single bit from a yes–no question, by exploiting the correlations of an intelligent prior: an answer over here settles expectations over there. This ability to extrapolate is codified here as *leverage*.

1 A thought experiment

Let us start with a small thought experiment, by introducing *Werner the learner* (Figure 1).

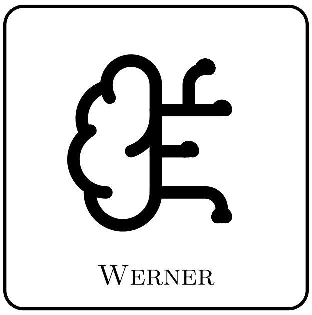


Figure 1: Werner the learner.

In this work, we model learning as the increased ability to give the correct answer to questions, as more of the ground truth is revealed. It is Werner's job to learn a pattern. The most basic shape of a pattern is a *binary map*, and the most basic binary map is a map from one bit to one bit.

The cast of the experiment:

- a **question** $q \in Q = \{0, 1\}$;
- an **answer** $a \in A = \{0, 1\}$;
- the **truth** $\psi : Q \to A$, the one fixed map nature uses to answer questions. ψ is unknown to Werner. By definition then, $\psi(q) = a$.

There are exactly four candidate maps ϕ_j that ψ could be (Table 1): each of the two inputs can be sent to either of the two outputs, independently. We index them by reading the truth table $\phi_j(1)\,\phi_j(0)$ as a binary numeral: that number *is* j.

j	$\phi_j(1)$	$\phi_j(0)$	name
0	0	0	constant 0
1	0	1	NOT ($\neg q$)
2	1	0	identity (q)
3	1	1	constant 1

Table 1: The four binary maps from one bit to one bit.

Now, if Werner is unintelligent, he has no preconceived notion of which maps are more useful, more likely, or somehow more logical than others. In practice, we would have to tell Werner the answer a to every single question q, before he has learned the whole pattern.

We model this with a **uniform prior**: writing $p_j = P(\psi = \phi_j)$ for Werner's belief, every candidate map carries the same weight.

EXAMPLE

For the four maps at hand:

$$p_j = \tfrac{1}{4}, \qquad j = 0, 1, 2, 3. \tag{1}$$

What does this belief predict? For each question q separately, we can tabulate the probability of each possible answer a: it is the total belief in the maps that would answer a to q,

$$M_{a,q} \;=\; \sum_{j\,:\,\phi_j(q)=a} p_j. \tag{2}$$

Arranged with one row per answer ($a = 0, 1$ from the top) and one column per question ($q = 1, 0$ from the left, descending like the digits of the numeral convention), these numbers form the matrix M - each column a probability distribution over answers, so M is column-stochastic.

EXAMPLE

Under the uniform prior, two of the four maps answer 0 and two answer 1, at either question:

$$M \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & M_{0,1} & M_{0,0} \\ \mathbf{1} & M_{1,1} & M_{1,0} \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0+p_1 & p_0+p_2 \\ \mathbf{1} & p_2+p_3 & p_1+p_3 \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{3}$$

Every column is a fair coin: the unintelligent Werner predicts nothing about any answer before it is revealed.

Next, Werner gets to predict the answer to the question $q = 0$. By his own belief it is a coin toss: $M_{0,0} = M_{1,0} = \frac{1}{2}$. Let us say the answer comes back $a = 1$. The **surprisal** of this answer, to Werner, is 1 bit, because the probability he assigned to it was one half.

In general, when an event a belief holds with probability P actually occurs, its surprisal is

$$s \;=\; -\log_2 P \quad \text{bits,} \tag{4}$$

here $s = -\log_2 M_{a,q}$ for answer a arriving at question q. Surprisal measures how much the received information deviates from what was already expected: an answer held certain ($P = 1$) carries 0 bits — nothing new; a coin toss carries exactly 1 bit; and the surprisal grows without bound as the answer gets more unexpected, $P \to 0$. It is additive: the surprisal of two independent events occurring is the sum of their surprisals, just as their probabilities multiply.

Moreover, Werner can use a process of elimination — an extreme form of Bayes' rule — to update his prior into a posterior. Semantically: every candidate map that would have answered the question wrongly is now *ruled out*, not merely disfavored.

EXAMPLE

The truth ψ answered 1 at $q = 0$, and a map with $\phi_j(0) = 0$ cannot be ψ. This kills the constant-0 map ($j = 0$) and the identity ($j = 2$). The two survivors, NOT ($j = 1$) and constant 1 ($j = 3$), both predicted the observed answer, so this round of evidence does not distinguish between them at all (Table 2).

j	$\phi_j(1)$	$\phi_j(0)$	name
~~0~~	~~0~~	~~**0**~~	~~constant 0~~
1	0	1	NOT ($\neg q$)
~~2~~	~~1~~	~~**0**~~	~~identity (q)~~
3	1	1	constant 1

Table 2: The candidate maps after the answer $\psi(0) = 1$: the two maps with $\phi_j(0) = 0$ are ruled out.

The surviving ratio of believabilities must be preserved, and the only update that does so is to renormalize the surviving weights to sum to one. (This is Bayes' rule with a 0/1 likelihood: $P(\psi = \phi_j \mid a) \propto P(a \mid \phi_j)\, p_j$, and $P(a \mid \phi_j)$ is 1 for a map that answers a and 0 for a map that does not.)

EXAMPLE

The posterior is

$$p' \;=\; \left(0,\ \tfrac{1}{2},\ 0,\ \tfrac{1}{2}\right). \tag{5}$$

Constructing the new answer matrix from p' by the same recipe as before:

$$M' \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p'_0 + p'_1 & p'_0 + p'_2 \\ \mathbf{1} & p'_2 + p'_3 & p'_1 + p'_3 \end{array} \;=\; \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & 0 \\ \mathbf{1} & \frac{1}{2} & 1 \end{array}. \tag{6}$$

So indeed, Werner knows the answer to $q = 0$ — that column has collapsed onto $a = 1$ — and he is still as agnostic as possible about the other answer: the $q = 1$ column remains a fair coin. He gained 1 bit of surprisal information, and cleaned up his table to the tune of 1 bit: the $q = 0$ column went from a full bit of answer-uncertainty to none, while the $q = 1$ column is unchanged.

After the next question, $q = 1$, he learns $a = 1$ — again a coin toss beforehand ($M'_{1,1} = \frac{1}{2}$), so again worth 1 bit of surprisal. Elimination now rules out NOT ($\phi_1(1) =$

0), leaving a single survivor:

$$p'' = \big(0,\ 0,\ 0,\ 1\big), \qquad M'' = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 0 & 0 \\ \mathbf{1} & 1 & 1 \end{array}. \tag{7}$$

So this is the constant map: $\psi = \phi_3$. Werner is done. Every column is one-hot, and he will be able to answer all questions correctly from now on.

2 Intelligent priors

Now, let us assume that Werner knows *something* about the world before he starts.

EXAMPLE

Let us say Werner knows the map is not constant. His belief then spreads evenly over the two remaining candidates, NOT ($j = 1$) and the identity ($j = 2$):

$$p = \big(0,\ \tfrac{1}{2},\ \tfrac{1}{2},\ 0\big). \tag{8}$$

Is he any better at predicting $\psi(0)$? Constructing M by the usual recipe:

$$M = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0 + p_1 & p_0 + p_2 \\ \mathbf{1} & p_2 + p_3 & p_1 + p_3 \end{array} = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{9}$$

He is still just as bad as before: every column a fair coin.

So what has all this veteran wisdom about the nature of patterns in the world bought him? His increased ability for *inference*.

EXAMPLE

Ask $q = 0$ again, and let the answer again come back $a = 1$ — again a coin toss beforehand, so again worth exactly 1 bit of surprisal. Elimination rules out the identity ($\phi_2(0) = 0$), and the lone survivor is NOT:

$$p' = \big(0,\ 1,\ 0,\ 0\big), \qquad M' = \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1 & 0 \\ \mathbf{1} & 0 & 1 \end{array}. \tag{10}$$

This time *both* columns collapse at once. Werner received the same single bit of surprisal as before, but cleaned up 2 bits of table: the answer at $q = 0$ he observed, and the answer at $q = 1$ he *deduced*, without ever asking — ruling out the constants means $\phi(1) \neq \phi(0)$, so his prior held the two answers perfectly anticorrelated, and settling one settles the other. He is done after one question instead of two.

We say the information from the first answer was **leveraged**. We will make this more precise in the upcoming sections.

2.1 A one-parameter world

For more structure, let us relax the claim that there are *no* constant maps. Instead, we create a one-parameter world where the prior probability that the map is constant is p, spread evenly within each class (Table 3):

j	$\phi_j(1)$	$\phi_j(0)$	name	p_j
0	0	0	constant 0	$p/2$
1	0	1	NOT ($\neg q$)	$(1-p)/2$
2	1	0	identity (q)	$(1-p)/2$
3	1	1	constant 1	$p/2$

Table 3: The one-parameter prior: constant with probability p.

In vector form, and with the answer matrix constructed by the usual recipe:

$$\big(p_0,\ p_1,\ p_2,\ p_3\big) \ = \ \tfrac{1}{2}\big(p,\ 1-p,\ 1-p,\ p\big), \qquad M \ = \ \begin{array}{c|cc} \mathbf{a\backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}, \tag{11}$$

fair coins in every column, for every value of p.

How much is there to learn, in total? We account for the table as a whole by declaring entropy *extensive* over the columns — the answer-uncertainties of the columns simply add:

$$H(M) \ := \ \sum_q H\big(M_{\cdot,q}\big), \qquad H\big(M_{\cdot,q}\big) \ := \ -\sum_a M_{a,q}\,\log_2 M_{a,q}. \tag{12}$$

Here both columns are fair coins carrying 1 bit each, so the table starts at its maximum: $H(M) = 2$ bits.

Now let us ask a question. Again we could use $q = 0$; by the symmetry of the prior it actually does not matter for this argument. And we are now going to talk about *expected* values, instead of realized ones: averages over the possible maps, denoted $\langle\,\cdot\,\rangle$. Since we assume our prior is *a priori* correct — nature draws the truth from the very distribution Werner believes — the average over ψ simply zips with the weights: $\langle X\rangle = \sum_j p_j\, X(\psi{=}\phi_j)$.

How much surprisal should Werner expect from asking a question q? The answer comes back a exactly when the truth answers a, which under the zipped average happens with probability $\sum_{j:\phi_j(q)=a} p_j = M_{a,q}$ — the very probability Werner assigned to it — and when it does, it costs him $-\log_2 M_{a,q}$ of surprisal. Grouping the average over maps by the answer each map gives:

$$\langle s_q\rangle \ = \ \sum_j p_j\,\big(-\log_2 M_{\phi_j(q),\,q}\big) \ = \ \sum_a \underbrace{\sum_{j\,:\,\phi_j(q)=a} p_j}_{M_{a,q}}\,\big(-\log_2 M_{a,q}\big) \ = \ \sum_a M_{a,q}\,\big(-\log_2 M_{a,q}\big). \tag{13}$$

The right-hand side is exactly the column entropy $H(M_{\cdot,q})$ of equation (12): **the expected surprisal of a question is the entropy of its column**. Each answer occurs with exactly the probability Werner assigned to it, so his average surprisal is his own uncertainty about that answer — an identity that will do a great deal of work in what follows.

For a two-answer column, this entropy is a function of a single number, the probability x of one of the two answers: the **binary entropy function**

$$h(x) \ := \ -x\log_2 x \ - \ (1-x)\log_2(1-x), \tag{14}$$

zero at $x \in \{0,1\}$ (a settled answer) and maximal, 1 bit, at the fair coin $x = \frac{1}{2}$. For the first question here the column is a fair coin, so

$$\langle s_1 \rangle \;=\; H\big(M_{\cdot,0}\big) \;=\; h\big(\tfrac{1}{2}\big) \;=\; 1 \text{ bit}, \tag{15}$$

whatever the value of p.

And the other column is now cleaned up. Breaking the update down over the possible maps: with probability $\frac{1}{2}$ the answer is 1 (truth is NOT or constant 1), with probability $\frac{1}{2}$ it is 0 (truth is FALSE or the identity), and

$$M' \;=\; \underbrace{\begin{array}{c|cc} \mathbf{a \backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1-p & 0 \\ \mathbf{1} & p & 1 \end{array}}_{a_1=1,\ \text{prob } 1/2} \quad \text{or} \quad \underbrace{\begin{array}{c|cc} \mathbf{a \backslash q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p & 1 \\ \mathbf{1} & 1-p & 0 \end{array}}_{a_1=0,\ \text{prob } 1/2}. \tag{16}$$

Either way, the asked column is one-hot and the unasked column is a coin of bias p: what survives on $q = 1$ is precisely the class uncertainty — constant or not — and nothing else. The total entropy of the table is now

$$H(M') \;=\; 0 + h(p) \;=\; h(p), \tag{17}$$

a reduction of $2 - h(p)$ bits, purchased with a single bit of surprisal. Define the **leverage after one question** as the ratio of table entropy destroyed to surprisal received:

$$L_1 \;:=\; \frac{2 - h(p)}{1} \;=\; 2 - h(p) \;\in\; [1,2]. \tag{18}$$

The expected surprisal on the second question is, once more, the entropy of its column,

$$\langle s_2 \rangle \;=\; h(p) \;\leq\; 1 \text{ bit}, \tag{19}$$

after which the table is empty. Cumulatively over the two questions, $1 + h(p)$ bits of received information reduced the table by the full 2 bits it started with, so the **cumulative leverage** after two questions is

$$L_2 \;:=\; \frac{2}{1 + h(p)} \;\in\; [1,2]. \tag{20}$$

So our knowledge about the world has allowed us to be more certain after our inference. Set $p = \frac{1}{2}$, and the prior is the uniform one of Section 1: $h = 1$, $L_1 = L_2 = 1$, and nothing is leveraged at all. As p skews further from $\frac{1}{2}$, the second question carries less and less expected information, and Werner's potential for learning from it goes down with it. The column it could clean up is nearly settled already. In the limit $p = 0$, we recover the non-constant world from the start of Section 2. There is no surprisal and no improvement to M left in the second question at all: it contributes nothing to denominator or enumerator, and $L_1 = L_2$. Figure 2 traces all three curves between these extremes.

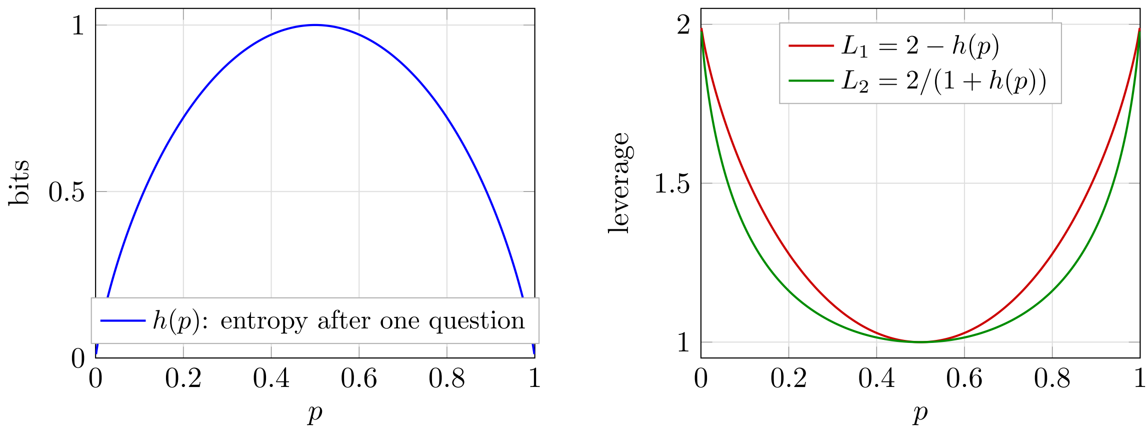


Figure 2: The one-parameter world as a function of the constant-map probability p. Left: the table entropy $h(p)$ remaining after one question, in bits. Right: the one-question leverage L_1 and the cumulative leverage L_2 after both questions. At $p = \frac{1}{2}$ (the uniform prior) nothing is leveraged; toward the deterministic-class extremes every answer doubles its value.

In general, in a larger world, if there are d questions to disagree on between the two maps, the reader can convince themselves that the limit as $p \to 1$ is $L_1 = d$.

3 The general case

Now let us give the definitions in full generality. Questions are bit strings of length n, and answers are bit strings of length m:

$$q \in Q = \{0,1\}^n, \qquad a \in A = \{0,1\}^m, \tag{21}$$

so there are 2^n distinct questions and 2^m possible answers to each. As before, we read bit strings as the numbers they spell, using the two interchangeably: $q \in \{0, \dots, 2^n{-}1\}$ and $a \in \{0, \dots, 2^m{-}1\}$. The truth $\psi : Q \to A$ is one of the candidate maps ϕ_j; since each of the 2^n inputs can be sent to any of the 2^m outputs independently, there are

$$N \;=\; \left(2^m\right)^{2^n} \;=\; 2^{m\,2^n} \tag{22}$$

of them, and the index j is again the truth table read as a numeral, now with 2^n digits in base 2^m: $j = \sum_q \phi_j(q)\,(2^m)^q$, running from the constant-0 map ($j = 0$) to the constant-($2^m{-}1$) map ($j = N{-}1$). Werner's belief is a distribution over all of them, $\sum_{j=0}^{N-1} p_j = 1$.

EXAMPLE

The example we will use to guide the general case is $n = 2$, $m = 1$, which has $N = 2^{1\cdot 4} = 16$ maps: every Boolean function of the two input bits b_1, b_0, with $q = b_1 b_0$ read as an ordinary binary number, $q = 2b_1 + b_0$. Each ϕ_j in Table 4 is shown as its truth table, printed in descending question order $(11, 10, 01, 00)$ so that the answer to question q occupies the digit of weight 2^q. The printed string is exactly j in binary (e.g. AND, which answers 1 only to $q = 11$, reads 1000, so it is $j = 8$). The prior p_j alongside is conjured for illustration.

j	$\phi_j(q)$ 11	10	01	00	name	p_j
0	0	0	0	0	FALSE (constant 0)	0.394
1	0	0	0	1	$\neg(b_1 \vee b_0)$ (NOR)	0.010
2	0	0	1	0	$\neg b_1 \wedge b_0$	0.010
3	0	0	1	1	$\neg b_1$	0.010
4	0	1	0	0	$b_1 \wedge \neg b_0$	0.131
5	0	1	0	1	$\neg b_0$	0.010
6	0	1	1	0	$b_1 \oplus b_0$ (XOR)	0.010
7	0	1	1	1	$\neg(b_1 \wedge b_0)$ (NAND)	0.010
8	1	0	0	0	$b_1 \wedge b_0$ (AND)	0.213
9	1	0	0	1	$b_1 = b_0$ (XNOR)	0.092
10	1	0	1	0	b_0 (projection)	0.040
11	1	0	1	1	$b_1 \to b_0$ (IF THEN)	0.010
12	1	1	0	0	b_1 (projection)	0.010
13	1	1	0	1	$b_0 \to b_1$ (IF THEN)	0.030
14	1	1	1	0	$b_1 \vee b_0$ (OR)	0.010
15	1	1	1	1	TRUE (constant 1)	0.010

Table 4: The guiding example: all 16 maps of the $n = 2$, $m = 1$ hypothesis space, with a contrived prior ($\sum_j p_j = 1.000$).

The answer matrix is defined by the same rule as before — the total belief in the maps that would answer a to q:

$$M_{a,q} = \sum_{j\,:\,\phi_j(q)=a} p_j, \tag{23}$$

now a $2^m \times 2^n$ matrix, still with each column a probability distribution over the possible answers: column-stochastic.

It is worth taking a moment to describe the constitution of this matrix. Because of the enumeration of j (the index *is* the truth table) the answer matrix has a natural structure: it is the contraction of the belief vector p with a fixed tensor with $\{0, 1\}$ entries E, summed along the map index,

$$M_{a,q} = \sum_{j=0}^{N-1} E_{a,q,j}\, p_j, \qquad E_{a,q,j} := \delta\big(\phi_j(q),\, a\big) = \delta\Big(\big\lfloor j \cdot 2^{-qm} \big\rfloor \bmod 2^m,\, a\Big). \tag{24}$$

The second form is pure arithmetic on the index: by the numeral convention, $\phi_j(q)$ is nothing but the q-th digit of j in base 2^m, so E never needs to be stored. Membership of j in the entry $M_{a,q}$ is read directly off the digits of j. This is a good thing, E is astronomically large for even moderate m and n. For $m = 1$ this reduces to: bit q of j equals a. The tensor E is moreover sparse and perfectly regular: each question-slice (q fixed) holds exactly N nonzero entries: every map answers exactly one a divided evenly over the 2^m answer rows, $N/2^m$ apiece. Within a slice the nonzero row follows the odometer pattern of a numeral system: runs of $(2^m)^q$ consecutive j share a digit, cycling through all 2^m rows with period $(2^m)^{q+1}$.

The odometer pattern gives every entry of M the same compact closed form: for $m = 1$, bit q of j equals 0 exactly when $j \bmod 2^{q+1} < 2^q$, and equals 1 otherwise.

EXAMPLE

Each entry of M collects the eight maps of Table 4 whose truth table carries the right digit. Filling the entries in — note the period halving column by column, with the outer columns written in their simplified forms, $j \bmod 16 < 8$ being simply $j < 8$ and $j \bmod 2 < 1$ simply "j even" — and then substituting the values of p_j:

$$
\begin{aligned}
M \;=\; & \begin{array}{c|cccc}
\mathbf{a\backslash q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\
\hline
\mathbf{0} & \sum_{j<8} p_j & \sum_{j \bmod 8<4} p_j & \sum_{j \bmod 4<2} p_j & \sum_{j \text{ even}} p_j \\
\mathbf{1} & \sum_{j\geq 8} p_j & \sum_{j \bmod 8\geq 4} p_j & \sum_{j \bmod 4\geq 2} p_j & \sum_{j \text{ odd}} p_j
\end{array} \\
\;=\; & \begin{array}{c|cccc}
\mathbf{a\backslash q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\
\hline
\mathbf{0} & 0.585 & 0.779 & 0.890 & 0.818 \\
\mathbf{1} & 0.415 & 0.221 & 0.110 & 0.182
\end{array} .
\end{aligned}
\tag{25}
$$

The update rule also carries over verbatim. When question q is asked and the answer $a = \psi(q)$ comes back, the process of elimination reads, in one formula,

$$
p'_j \;=\; \frac{\delta\big(\phi_j(q),\, a\big)\; p_j}{M_{a,q}} \;=\; \frac{E_{a,q,j}\; p_j}{M_{a,q}}. \tag{26}
$$

The Kronecker delta performs the elimination. Only maps that agree with a survive and the division renormalizes the survivors, whose total prior mass is exactly $M_{a,q}$ by equation (23). Note that one and the same number prices the answer and funds the update: the event arrives with the probability $M_{a,q}$, costs Werner $-\log_2 M_{a,q}$ bits of surprisal, and its belief mass becomes the new unit of normalization. Every update shrinks the support by 2^{-m}, radically thinning, especially for large m. The survivors live on a block-wave in j-space. The map determines the offset and frequency. The total set of survivors depends on the set of asked questions, not the order: updating commutes. This is illustrated in the example below.

EXAMPLE

Figure 3 makes the process visual for $n = m = 2$: a schematic prior over $N = 4^4 = 256$ maps, drawn as vertical lines with thickness representing weight. Each answer stops 3 out of every 4 surviving lines: maps whose truth table carry a wrong digit at the asked question. Renormalization inflates the weight of the rest, so the total probability is always unit. Four questions cut $256 \to 64 \to 16 \to 4 \to 1$, and the last line standing is the truth. Note the regularity of the cull: each answer keeps a single residue class of j, so the cadence of the survivors is regular.

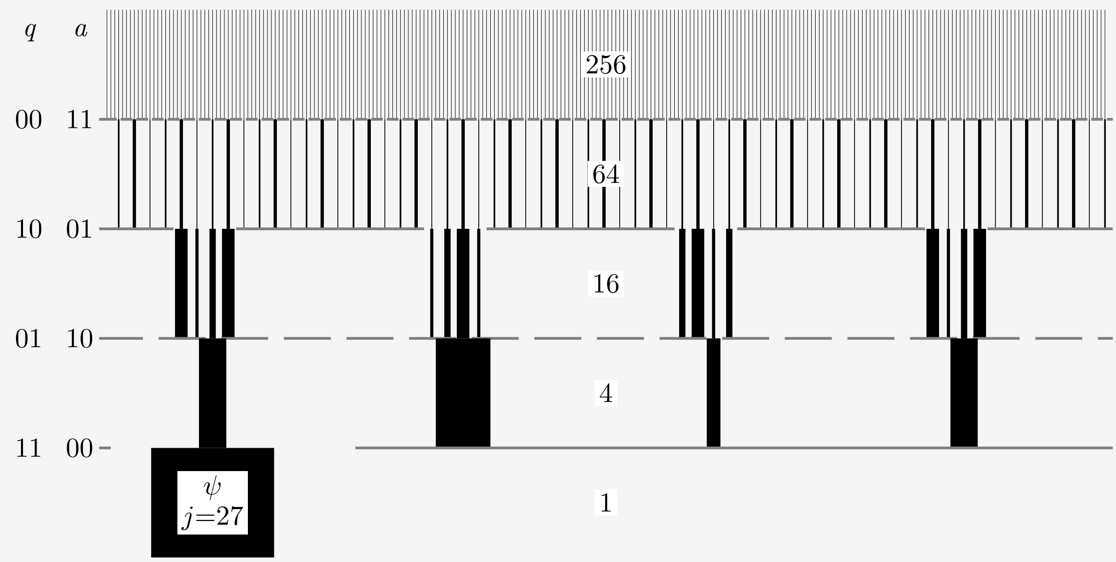


Figure 3: The update rule as a cull, for $n = m = 2$. All 256 candidate maps start as vertical lines under a schematic prior. The truth is the bitwise negation $\psi(q) = \neg q$ ($j = 27$), and the four questions are asked in the order $00, 10, 01, 11$. Each answer terminates the 3/4 of surviving maps whose truth-table digit at the asked question disagrees. Every survivor's thickness, proportional to its weight, is rescaled by the same renormalization factor, preserving their ratios. After four questions a single map remains. Interestingly, the first cull's survivors are evenly spaced because it's the smallest question. After the three lowest q, even spacing returns.

The total entropy of the table is extensive over the columns, exactly as in equation (12):

$$H(M) \;=\; \sum_{q\in Q} H\big(M_{\cdot,q}\big) \;=\; -\sum_{q\in Q}\sum_{a\in A} M_{a,q}\log_2 M_{a,q} \;\;\le\;\; m\,2^n, \tag{27}$$

each column carrying at most m bits (a uniform spread over its 2^m answers). The ceiling $m\,2^n$ is the size of the entire truth table in bits, and it is attained precisely by the uniform prior: maximal ignorance prices every answer at full cost.

The belief itself carries an entropy as well, and we give its gap to the table a name:

$$H(p) \;=\; -\sum_{j=0}^{N-1} p_j \log_2 p_j, \qquad C \;:=\; H(M) - H(p) \;\ge\; 0. \tag{28}$$

$H(p)$, Werner's uncertainty about the *identity* of the truth, obeys the same ceiling: a distribution over N maps has entropy at most $\log_2 N = m\,2^n$, attained again by the uniform prior. The information in the prior is not directly useful for prediction, but it is good to track.

Why is $C > 0$? The answers jointly determine the map, so $H(p)$ is the *joint* entropy of all 2^n answers, while $H(M)$ sums their *marginal* entropies and a joint entropy never exceeds the sum of its marginals. Hence $H(p) \le H(M)$, and the gap C, the **total correlation** of the answers, is nonnegative: knowledge stored in correlations *between* answers, invisible to every single column of M.

We choose this convention for $H(M)$ because in the ignorant case

1. $H(M)$ is the total amount of information we must collect from questions to settle the map
2. $H(M) = H(p)$, so the ignorance agrees in both bases, without correlations.

Therefore, the scaling is sensible. How these quantities respond to an intelligent prior, is the main question of this work.

Let us watch these ledgers in motion by playing the guiding example through all four questions. Werner can compute two forecasts for every unasked question, in advance, from M alone: the expected surprisal $\langle s \rangle$, which by equation (13) is the entropy of that question's column, and the expected drop $\langle \Delta H(M) \rangle$ of the *whole* table if that question is asked next. This is the average over possible a, weighted by their probability (according to M):

$$\langle s \rangle(q) \;=\; \sum_a M_{a,q}\left(-\log_2 M_{a,q}\right), \tag{29}$$

$$\langle \Delta H(M) \rangle(q) \;=\; \sum_a M_{a,q}\left[H(M) - H\big(M^{(q,a)}\big)\right], \tag{30}$$

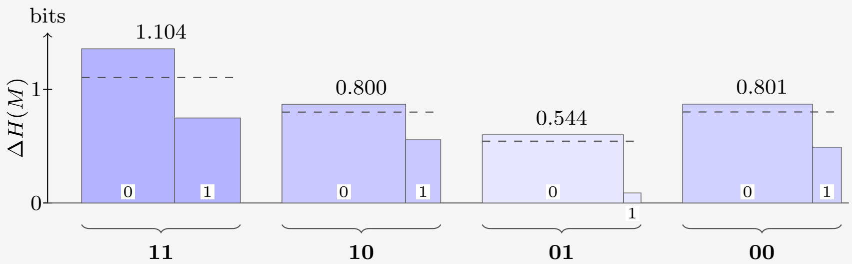
where $M^{(q,a)}$ is the table after the update rule (26) processes answer a to question q. We know the column of q will drop $\langle s \rangle(q)$ in entropy. But other columns will also respond. The excess of the second over the first is the expected transfer to unasked columns, and it can only be paid out of correlations. We append both forecasts as extra rows under M.

EXAMPLE

Werner plays greedily: always asking the question with the largest forecast. The truth is AND ($\psi = \phi_8$).

Question 1. The starting state:

a\q	**11**	**10**	**01**	**00**
0	0.585	0.779	0.890	0.818
1	0.415	0.221	0.110	0.182
$\langle s \rangle$	0.979	0.762	0.500	0.684
$\langle \Delta H(M) \rangle$	1.104	0.800	0.544	0.801



To see where the forecast entries come from, take $q = 11$: the answer $a = 0$ arrives with probability $M_{0,11} = 0.585$ and would drop $H(M)$ from 2.925 to 1.569; the answer $a = 1$ arrives with probability 0.415 and drops it only to 2.178, so

$$\langle s \rangle(11) = 0.585\,(0.773) + 0.415\,(1.269) = 0.979,$$
$$\langle \Delta H(M) \rangle(11) = 0.585\,(1.357) + 0.415\,(0.748) = 1.104.$$

Both forecast rows point to $q = 11$: it is the closest thing to a coin on offer ($\langle s \rangle = 0.979$) and also moves the whole table most, by a wide margin. Werner asks it, and nature answers $a = 1$ (probability 0.415, surprisal $s_1 = 1.269$). Eight maps die:

j	$\phi_j(11)$	name	p_j	p'_j
~~0~~	~~0~~	~~FALSE (constant 0)~~	~~0.394~~	~~—~~
~~1~~	~~0~~	~~$\neg(b_1 \vee b_0)$ (NOR)~~	~~0.010~~	~~—~~
~~2~~	~~0~~	~~$\neg b_1 \wedge b_0$~~	~~0.010~~	~~—~~
~~3~~	~~0~~	~~$\neg b_1$~~	~~0.010~~	~~—~~
~~4~~	~~0~~	~~$b_1 \wedge \neg b_0$~~	~~0.131~~	~~—~~
~~5~~	~~0~~	~~$\neg b_0$~~	~~0.010~~	~~—~~
~~6~~	~~0~~	~~$b_1 \oplus b_0$ (XOR)~~	~~0.010~~	~~—~~
~~7~~	~~0~~	~~$\neg(b_1 \wedge b_0)$ (NAND)~~	~~0.010~~	~~—~~
8	1	$b_1 \wedge b_0$ (AND)	0.213	0.513
9	1	$b_1 = b_0$ (XNOR)	0.092	0.222
10	1	b_0 (projection)	0.040	0.096
11	1	$b_1 \to b_0$ (IF THEN)	0.010	0.024
12	1	b_1 (projection)	0.010	0.024
13	1	$b_0 \to b_1$ (IF THEN)	0.030	0.072
14	1	$b_1 \vee b_0$ (OR)	0.010	0.024
15	1	TRUE (constant 1)	0.010	0.024

The ledger:

$$H(M)\colon 2.925 \to 2.178, \qquad H(p)\colon 2.707 \to 2.093, \qquad C\colon 0.218 \to 0.085.$$

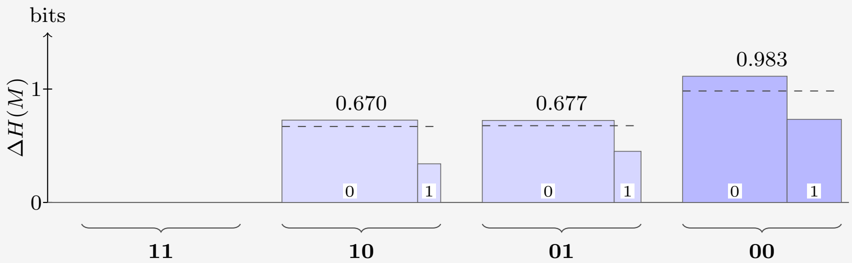
Werner received 1.269 bits but destroyed only 0.748 bits of table. Correlations point the right way only on average, not on every draw. In this case, Werner got the much less informative branch, and paid the premium price for it. The destroyed-per-received ratio of this round is 0.589: well below 1, anti-leverage.

Question 1 is a reminder that the forecast rows are *expectations*: they hold on average over many repetitions of the experiment, and only if the world really does draw its truth ψ from Werner's prior. On any single run the realized numbers are free to deviate.

EXAMPLE

Question 2. The state after the collapse, with fresh forecasts:

a\q	**11**	**10**	**01**	**00**
0	0	0.855	0.831	0.658
1	1	0.145	0.169	0.342
$\langle s \rangle$	—	0.596	0.655	0.927
$\langle \Delta H(M) \rangle$	—	0.670	0.677	0.983



Greedy picks $q = 00$. Nature answers $a = 0$ — the answer Werner favored (probability

0.658), so the surprisal is only $s_2 = 0.604$ bits. Four of the eight die:

j	$\phi_j(00)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.513	0.780
~~9~~	~~1~~	~~$b_1 = b_0$ (XNOR)~~	~~0.222~~	~~—~~
10	0	b_0 (projection)	0.096	0.147
~~11~~	~~1~~	~~$b_1 \to b_0$ (IF THEN)~~	~~0.024~~	~~—~~
12	0	b_1 (projection)	0.024	0.037
~~13~~	~~1~~	~~$b_0 \to b_1$ (IF THEN)~~	~~0.072~~	~~—~~
14	0	$b_1 \vee b_0$ (OR)	0.024	0.037
~~15~~	~~1~~	~~TRUE (constant 1)~~	~~0.024~~	~~—~~

The ledger:

$$H(M)\colon 2.178 \to 1.065, \qquad H(p)\colon 2.093 \to 1.035, \qquad C\colon 0.085 \to 0.030.$$

This time 1.113 bits are destroyed for 0.604 received. Where did the extra 0.509 come from? Partly from the favored answer arriving — the belief dropped 1.058 against only 0.604 paid — and partly from the store: C fell by 0.055. Step ratio 1.842. Cumulatively, however, 1.860 destroyed per 1.873 received gives a leverage of 0.993 after 2 steps: still a hair below 1. Even a strong second round has not quite repaid the expensive shock of question 1.

Why track C, when its step-by-step changes visibly do not match the amounts deduced? Because on a single realized run the deduced part has *two* sources, and C is the only quantity that separates them. Writing Δ for the realized drops in one step,

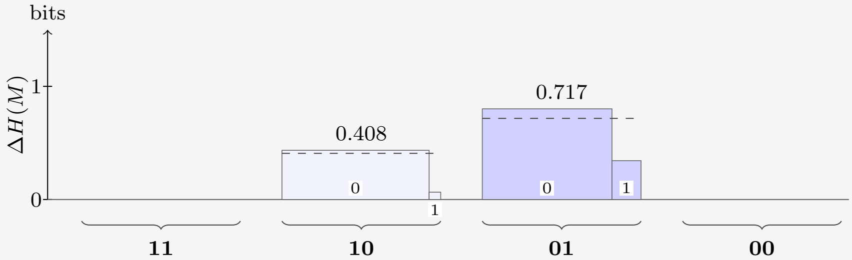
$$\underbrace{\Delta H(M) - s}_{\text{deduced}} = \underbrace{\big[\Delta H(p) - s\big]}_{\text{windfall}} + \underbrace{\big[-\Delta C\big]}_{\text{tower}}, \tag{31}$$

an identity, since $H(M) = H(p) + C$ at every stage. The windfall is luck: the realized surprisal undershooting (or overshooting) the realized drop in belief entropy: zero in expectation. The tower term is structure: a genuine withdrawal from the correlation store, and the channel with a global guarantee. Over a complete run its withdrawals sum to exactly the initial C, while the windfalls average away.

EXAMPLE

Question 3. Two questions remain:

a\q	**11**	**10**	**01**	**00**
0	0	0.927	0.817	1
1	1	0.073	0.183	0
$\langle s \rangle$	—	0.378	0.687	—
$\langle \Delta H(M) \rangle$	—	0.408	0.717	—



Greedy picks $q = 01$; nature answers $a = 0$ (probability 0.817, surprisal $s_3 = 0.292$). Two die:

j	$\phi_j(01)$	name	p_j	p'_j	
8	0	$b_1 \wedge b_0$ (AND)	0.780	0.955	
~~10~~	~~1~~	~~b_0 (projection)~~	~~0.147~~	~~—~~	
12	0	b_1 (projection)	0.037	0.045	
~~14~~	~~1~~	~~$b_1 \vee b_0$ (OR)~~	~~0.037~~	~~—~~	

The ledger:

$$H(M)\colon 1.065 \to 0.264, \qquad H(p)\colon 1.035 \to 0.264, \qquad C\colon 0.030 \to 0.$$

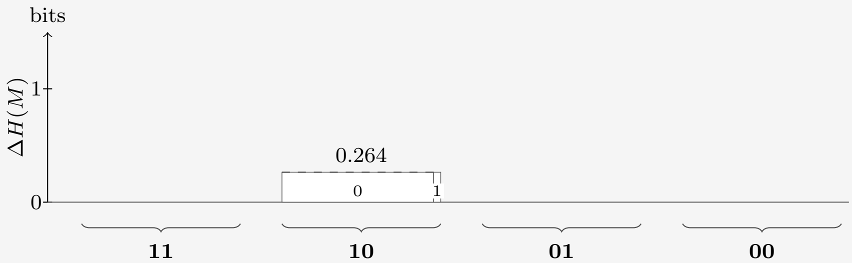
Again a bargain: 0.801 destroyed for a mere 0.292 received, and the cumulative ratio finally clears unity, at 1.229. And note that C has dropped to zero.

C measures correlations that are not visible from M. If there's only one unknown column, there's nothing to infer. Put another way, usually, the binary maps are an overcomplete basis for M, because their number $N \gg \dim(M) = 2^{m+n}$. That's not the case when the number of free parameters has been reduced by learning to 1 each.

EXAMPLE

Question 4. One question, one heavily loaded coin:

a\q	**11**	**10**	**01**	**00**
0	0	0.955	1	1
1	1	0.045	0	0
$\langle s \rangle$	—	0.264	—	—
$\langle \Delta H(M) \rangle$	—	0.264	—	—



The two forecasts now coincide: with $C = 0$ there is no transfer left to hope for, and the last question can only be observed, not leveraged. Nature answers $a = 0$ (probability 0.955, surprisal $s_4 = 0.066$):

j	$\phi_j(10)$	name	p_j	p'_j	
8	0	$b_1 \wedge b_0$ (AND)	0.955	1.000	
~~12~~	~~1~~	~~b_1 (projection)~~	~~0.045~~	~~—~~	

The ledger closes:

$$H(M)\colon 0.264 \to 0, \qquad H(p)\colon 0.264 \to 0, \qquad C\colon 0 \to 0.$$

The truth is AND. The step destroyed 0.264 for 0.066 received (ratio 3.990) — not deduction this time, but luck: the destroyed amount was fixed in advance (the column's full entropy), while the realized surprisal undershot it because the likelier answer arrived. In the split of equation (31): pure windfall, tower 0.

Cumulatively: 2.925 bits of table destroyed, all of it, for 2.231 bits of surprisal received: a run ratio of 1.311. Most of the realized 1.311 was luck, not intelligence. Figure 4 stacks the books after each question: what remains, plus what was received. Once it dips below the starting entropy, our prior has done some work.

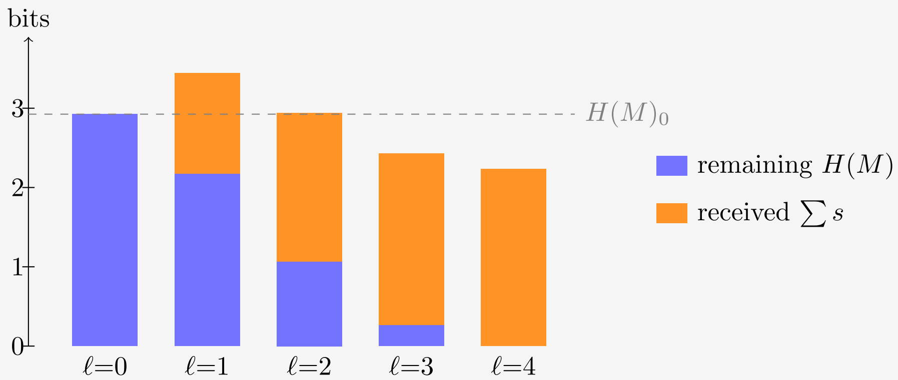


Figure 4: The realized run through the guiding example: remaining table entropy plus cumulative surprisal received, after each question, against the initial total (dashed). The gap between each stack and the line is the cumulative deduced part. At $\ell = 1$ the stack tops out 0.521 bits *above* the line — the negative deduction of question 1, plainly visible. At $\ell = 2$ the stack still grazes the line from above (0.013 bits); only from $\ell = 3$ on do the stacks fall short of it, as genuine deduction overtakes the early loss.

If the prior is good, meaning it actual models reality (the prevalence of maps), we expect graphs such as the one above to drop as questions come in. The ratio of table entropy destroyed per bit of surprisal received deserves a name.

The **leverage**. Run the protocol for k rounds: questions $q_1, \ldots, q_k$ are asked, the answers $a_i = \psi(q_i)$ are each processed by the update rule (26), producing the tables $M^{(i)}$, and round i costs the surprisal $s_i = -\log_2 M^{(i-1)}_{a_i, q_i}$. The cumulative leverage after k rounds is the table entropy destroyed per bit of surprisal received:

$$L_k \;:=\; \frac{H\big(M^{(0)}\big) - H\big(M^{(k)}\big)}{\sum_{i=1}^{k} s_i}\,. \tag{32}$$

A leverage of 1 is the uniform-prior baseline: every bit of table entropy destroyed was paid for at face value, one bit of surprisal each. Anything above 1 is deduction — table entropy removed without ever being observed.

4 Correlators

What kind of operator is the update rule (26)? In map space it is as simple as an operator gets: multiplication by a diagonal 0/1 mask, then renormalization. But there is a change of basis in which the same operation becomes a *convolution* with a two-term kernel, and in which the correlation store C stops being a single number and becomes visible coordinate by coordinate. The main advantage is that the effect of each update is well tabulated beforehand into a small number of targeted parameters.

This section builds that basis for $m = 1$; the final paragraph deals with general m.

First, re-encode each answer as a spin,

$$\sigma_q(\phi_j) \;:=\; (-1)^{\phi_j(q)} \;\in\; \{+1, -1\}, \tag{33}$$

so answer $0 \mapsto +1$ and $1 \mapsto -1$. A pure relabeling, but it buys the one algebraic fact everything below turns on: $\sigma_q^2 = 1$.

The **Walsh–Hadamard transform** is built from a single matrix. For one bit,

$$W \;=\; \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad W_{c,b} \;=\; (-1)^{c\,b}, \quad c, b \in \{0, 1\}. \tag{34}$$

EXAMPLE

Acting on a one-bit distribution,

$$W\begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \;=\; \begin{pmatrix} p_0 + p_1 \\ p_0 - p_1 \end{pmatrix} \;=\; \begin{pmatrix} 1 \\ \langle\sigma\rangle \end{pmatrix}:$$

row $c = 0$ computes the normalization, row $c = 1$ the spin bias. Distribution in, correlators out.

The normalization riding along is a redundancy. The original distribution also uses one variable more than its degrees of freedom.

EXAMPLE

Tensoring two copies (the $n = 1$ case) does the same for two bits at once:

$$W \otimes W \;=\; \begin{pmatrix} W & W \\ W & -W \end{pmatrix} \;=\; \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \quad \begin{array}{l} \leftarrow c = 00\!: \;\text{sums to } 1 \\ \leftarrow c = 01\!: \;\langle\sigma_0\rangle \\ \leftarrow c = 10\!: \;\langle\sigma_1\rangle \\ \leftarrow c = 11\!: \;\langle\sigma_1\sigma_0\rangle \end{array} \tag{35}$$

The Kronecker product multiplies signs, so for any number l of tensor factors the entry at row c, column j is

$$\left(W^{\otimes l}\right)_{c,j} \;=\; \prod_{b=0}^{l-1} (-1)^{c_b\, j_b} \;=\; (-1)^{\sum_b c_b\, j_b}, \tag{36}$$

with c_b, j_b the b-th bits of the row and column indices; every further tensor factor appends one more bit.

For $m = 1$ a map *is* its truth table of 2^n bits — the numeral j — so Werner's belief is a vector of length $N = 2^{2^n}$, and one transform computes every correlator at once:

$$\chi \;=\; W^{\otimes 2^n} p, \qquad \chi_S \;=\; \left\langle \prod_{q \in S} \sigma_q \right\rangle \;=\; \sum_{j=0}^{N-1} (-1)^{\sum_{q \in S} \phi_j(q)}\, p_j. \tag{37}$$

Each row computes the **correlator** of one subset $S \subseteq Q$ of questions. In the subscript we may encode the set as its *mask*: written in the same descending question order as everywhere else, bit q of the subscript records whether $q \in S$. The row index of the matrix, read as a numeral, is the mask.

EXAMPLE

For $n = 2$, the question order is $(11, 10, 01, 00)$ so χ_{0110} is the correlator of $S = \{10, 01\}$.

the sixteen rows of $W^{\otimes 4}$ probe the sixteen subsets of $Q = \{11, 10, 01, 00\}$. A few entries of the dictionary:

mask	set S	correlator
0000	$\emptyset$	$\chi_{0000} = 1$ (normalization)
0001	$\{00\}$	$\chi_{0001} = \langle \sigma_{00} \rangle$
1000	$\{11\}$	$\chi_{1000} = \langle \sigma_{11} \rangle$
0110	$\{10, 01\}$	$\chi_{0110} = \langle \sigma_{10}\, \sigma_{01} \rangle$
1111	$\{11, 10, 01, 00\}$	$\chi_{1111} = \langle \sigma_{11}\, \sigma_{10}\, \sigma_{01}\, \sigma_{00} \rangle$, the full parity

Sort the N correlators by $|S|$, the number of questions probed, into *shells*. Shell 0 is the normalization, $\chi_\emptyset = 1$, riding along as a correlator. Shell 1 is M in different clothes:

$$\chi_{\{q\}} \;=\; M_{0,q} - M_{1,q}, \qquad M_{a,q} \;=\; \tfrac{1}{2}\big(1 + (-1)^a\, \chi_{\{q\}}\big), \tag{38}$$

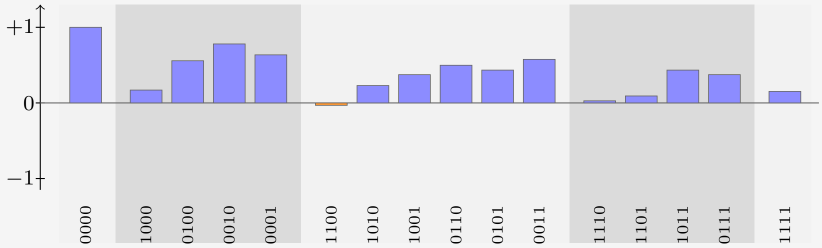
the bias of column q. For $m = 1$, we elegantly see the first shell correlators are the distance from fair coins.

Shell 2 holds the agreement biases of pairs of questions — knowledge M cannot see: two priors can share every column of M and differ here. Higher shells store subtler and subtler parity knowledge. Nothing is lost in the sorting: $W^2 = 2I$, so $(W^{\otimes 2^n})^2 = NI$ — the transform is its own inverse up to a factor N, and knowing every correlator is knowing the belief. The ledger also gives C a signature: a product prior ($C = 0$) factorizes every correlator, $\chi_S = \prod_{q \in S} \chi_{\{q\}}$, so its upper shells repeat what shell 1 already said; correlated knowledge is precisely *non-factorizing* correlators.

EXAMPLE

The prior of Table 4 transforms to the ledger

shell	correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = +0.170,\;\; \chi_{0100} = +0.558,\;\; \chi_{0010} = +0.780,\;\; \chi_{0001} = +0.636$
2	$\chi_{1100} = -0.032,\;\; \chi_{1010} = +0.230,\;\; \chi_{1001} = +0.374,$
	$\chi_{0110} = +0.498,\;\; \chi_{0101} = +0.434,\;\; \chi_{0011} = +0.576$
3	$\chi_{1110} = +0.028,\;\; \chi_{1101} = +0.092,\;\; \chi_{1011} = +0.434,\;\; \chi_{0111} = +0.374$
4	$\chi_{1111} = +0.152$



Shell 1 reproduces the columns of equation (25): $(1+0.170)/2 = 0.585 = M_{0,11}$, and likewise 0.779, 0.890, 0.818. The negative entry χ_{1100} says questions 11 and 10 disagree slightly more often than they agree; every other pair leans toward agreement. This is a reflection of the strong coherent 0-preference encoded in the prior. Answering 1 would give negative 1-shells.

Now look at the *rows* of $W^{\otimes 2^n}$ as functions of j. They are **block waves**. The observation mask is built of the same material:

$$\delta\big(\phi_j(q), a\big) \;=\; \frac{1}{2}\Big(1 + (-1)^{a+\phi_j(q)}\Big)\,, \tag{39}$$

a 0/1 block wave — one constant plus one Hadamard row. And multiplication and convolution swap under this transform exactly as under Fourier: multiplying two functions of j pointwise convolves their ledgers. And the observation mask's ledger has only two nonzero entries — at $\emptyset$ and at $\{q\}$ — so this convolution is no great sum: it combines each correlator with exactly one other.

The update rule, carried out entirely in the ledger, is therefore that two-term combination, followed by renormalization by the new shell 0. In index notation, learning $a = \psi(q)$ sends every set S to

$$\chi'_S \;=\; \frac{\chi_S + (-1)^a\, \chi_{S\triangle\{q\}}}{1 + (-1)^a\, \chi_{\{q\}}}, \tag{40}$$

where $S\triangle\{q\}$ is S with the membership of q toggled: q is added if absent, removed if present (on masks, bit q is flipped). The numerator is the monomial algebra of $\sigma_q^2 = 1$: multiplying $\prod_{r\in S}\sigma_r$ by σ_q toggles q's membership of the product, so every correlator is averaged with its **partner**: the correlator differing by q, signed by the answer. The denominator is the $S = \emptyset$ instance of the numerator, and by equation (38) it equals $2M_{a,q}$: the same number that priced the answer in equation (26) renormalizes the ledger.

EXAMPLE

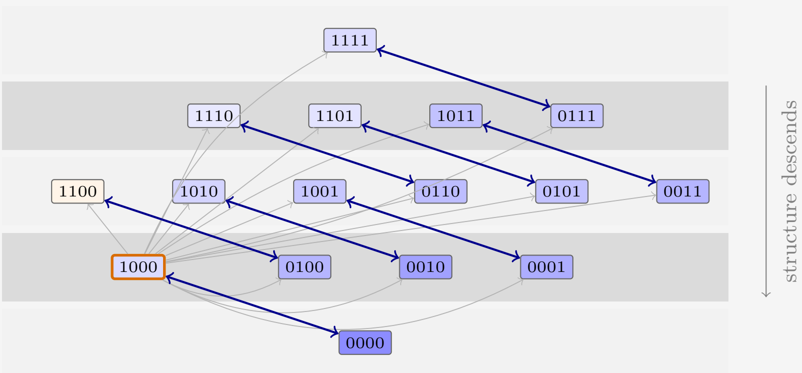


Figure 5: The update rule in the ledger, drawn for the guiding example's first question ($q = 11$). Every set is paired with the partner whose membership of 11 is toggled (arrows); each pair is averaged with the answer's sign and renormalized, per equation (40). The pair $\emptyset \leftrightarrow \{11\}$ at the orange-bordered asked set fixes the normalization and saturates the asked bias; each blue double arrow averages a pair into both of its members, and the net flow of structure is one shell down, toward M. The thin gray arrows from the asked q to every other pill remind that this sets the renormalization. The fill encodes each prior value to scale: darker is larger, orange is negative.

Read formula (40) by cases:

- $S = \emptyset$: partner $\{q\}$, giving $(1 + (-1)^a \chi_{\{q\}})/(1 + (-1)^a \chi_{\{q\}}) = 1$. Normalization is preserved.

- $S = \{q\}$, the asked question: partner $\emptyset$, giving $(\chi_{\{q\}} + (-1)^a)/(1 + (-1)^a \chi_{\{q\}}) = (-1)^a$. The bias saturates to certainty — the one-hot collapse of column q, derived rather than decreed.

- $S = \{q'\}$, an unasked question: the partner is the pair $\{q', q\}$, and one line of algebra puts the shift in terms of the *connected* correlator:

$$\chi'_{\{q'\}} - \chi_{\{q'\}} = \frac{(-1)^a \big(\chi_{\{q',q\}} - \chi_{\{q'\}}\, \chi_{\{q\}}\big)}{2M_{a,q}} = \frac{(-1)^a \operatorname{Cov}\big(\sigma_{q'}, \sigma_q\big)}{2M_{a,q}}. \tag{41}$$

  Unasked columns move in proportion to their covariance with the asked question: knowledge stored one shell up descends into the predictions. A product prior has zero covariance everywhere, so its unasked columns are frozen — the "$C = 0$ means no transfer" fact, now a one-line consequence.

- Higher shells: the same cascade. Each update marries shell-k sets containing q to shell-$(k-1)$ sets without it; whatever structure was stored jointly with q moves one shell down, toward M.

EXAMPLE

Question 1 of the walkthrough asked $q = 11$ and received $a = 1$: sign $(-1)^a = -1$, toggled set $\{11\}$, mask 1000. Equation (40) maps the ledger of the previous box to

shell	updated correlators
0	$\chi'_{0000} = 1$
1	$\chi'_{1000} = -\mathbf{1}$, $\chi'_{0100} = +0.711$, $\chi'_{0010} = +0.663$, $\chi'_{0001} = +0.316$
2	$\chi'_{1100} = -0.711$, $\chi'_{1010} = -0.663$, $\chi'_{1001} = -0.316$, $\chi'_{0110} = +0.566$, $\chi'_{0101} = +0.412$, $\chi'_{0011} = +0.171$
3	$\chi'_{1110} = -0.566$, $\chi'_{1101} = -0.412$, $\chi'_{1011} = -0.171$, $\chi'_{0111} = +0.267$
4	$\chi'_{1111} = -0.267$

The asked bias saturates: $\chi'_{1000} = (0.170 - 1)/(1 - 0.170) = -1$. The other singletons move by equation (41): for $q' = 10$ the covariance is $-0.032 - (0.558)(0.170) = -0.127$, so the shift is $(-1)(-0.127)/(2 \times 0.415) = +0.153$, landing on $+0.711$ — and indeed $(1 + 0.711)/2 = 0.855$, the column $M_{0,10}$ of the Question 2 box. Note also that every mask containing the asked bit is now just $(-1)^a$ times its partner ($\chi'_{1010} = -\chi'_{0010}$, etc.): with σ_{11} pinned, no correlator involving question 11 carries independent knowledge anymore.

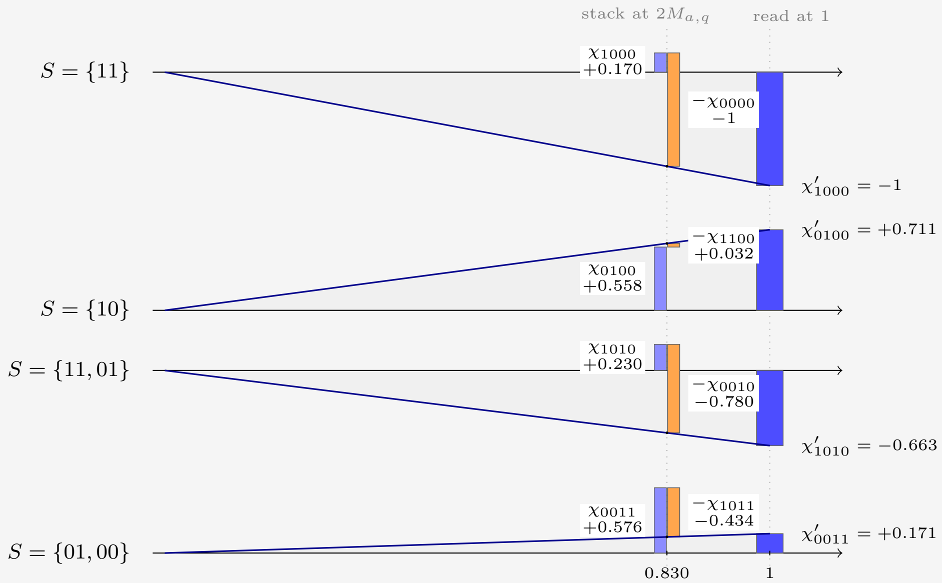


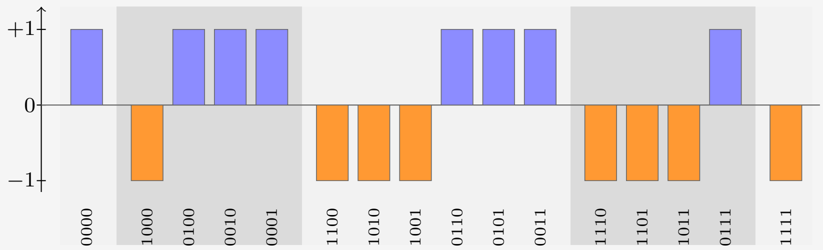
Figure 6: The update rule of equation (40), drawn for the guiding example's first update ($q = 11$, $a = 1$, so $(-1)^a = -1$ and toggled bit 1000). One row per set S. At $x = 2M_{a,q} = 1 + (-1)^a \chi_{1000} = 0.830$ stands the numerator as a waterfall: the bar of χ_S (blue), and beside it the partner's signed contribution $(-1)^a \chi_{S \triangle \{q\}}$ (orange), hanging from the level where the first bar ends. The two subscripts differ by the flipped bit. The ray from the origin through the waterfall's endpoint rescales it, by similar triangles, to $x = 1$, where its height is the updated correlator χ'_S.

As Werner moves through the questions, more correlators settle to ± 1. The 1-shell models the belief of the bias of each question. The 2-shell answers questions such as 'If the answer to $\psi(10) = 0$, then $\psi(01) = 1$'. And the higher orders are even more conditional, measuring belief about such statements, in the case of certain answers, etc. As we learn more, the conditions become satisfied or not, and the information becomes less nested.

EXAMPLE

The remaining questions of the run (00, 01, 10, all answered 0, so with sign +1) each saturate their own bit the same way, and since XOR toggles commute, the order does not matter — the commutativity of updating, seen again. After all four, the ledger is pure signs, $\chi_S = (-1)^{|S\cap\{11\}|}$ — -1 on every set containing 11, $+1$ on the rest: the transform of the point mass on $\psi = \phi_8$:

shell	final correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = -1,\ \ \chi_{0100} = +1,\ \ \chi_{0010} = +1,\ \ \chi_{0001} = +1$
2	$\chi_{1100} = -1,\ \ \chi_{1010} = -1,\ \ \chi_{1001} = -1,$
	$\chi_{0110} = +1,\ \ \chi_{0101} = +1,\ \ \chi_{0011} = +1$
3	$\chi_{1110} = -1,\ \ \chi_{1101} = -1,\ \ \chi_{1011} = -1,\ \ \chi_{0111} = +1$
4	$\chi_{1111} = -1$



4.1 General m

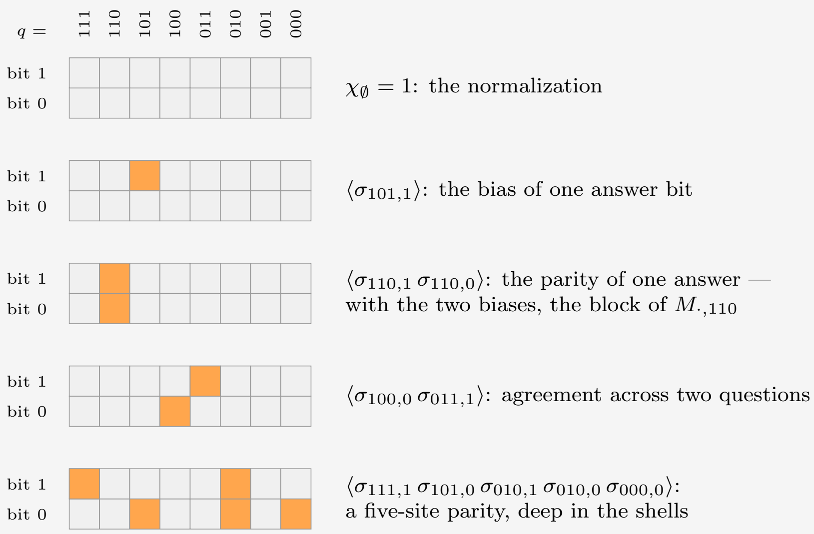
It turns out that taking $m > 1$ is more general but not much more instructive. With multiple output bits it is as if we have m a priori independent maps per question; they just happen always to be asked and answered simultaneously, in unison. Therefore we do not have to think about what one of them teaches us about another. It would mean we have more than 2 options every question, and therefore the factor by which every update thins the hypotheses is not 2 but 2^m. The guiding example helps because the maps are represented as binary numbers, which are more familiar.

For completeness, the main formulas. The truth table is now $m \cdot 2^n$ bits: one *site* (q, i) per bit i of each answer, each carrying a spin $\sigma_{q,i}(\phi_j) := (-1)^{\phi_j(q)_i}$, where $\phi_j(q)_i$ is the i-th bit of the answer. Correlators are indexed by sets S of sites — masks of $m2^n$ bits under the same numeral convention, m of them per question:

$$\chi_S = \Big\langle \prod_{(q,i)\in S} \sigma_{q,i} \Big\rangle = \sum_{j=0}^{N-1} (-1)^{\sum_{(q,i)\in S} \phi_j(q)_i}\, p_j. \tag{42}$$

EXAMPLE

For $n = 3$, $m = 2$ the sites form a 2×8 grid: one column per question, one row per bit of the answer. A set S is a pattern of hot cells in this grid, and χ_S multiplies the corresponding spins:



Any of the $2^{16} = 65536$ hot/cold patterns is a correlator, and sorting by the number of hot cells gives the shells. The mask of S is the grid read as a 16-bit numeral.

Write B_q for the block of q's m sites, and $a \cdot T := \sum_{i \in T} a_i$ for the parity of the answer a on a subset $T \subseteq B_q$. The answer matrix is read off the correlators supported on one block — 2^m of them per column:

$$M_{a,q} \;=\; \frac{1}{2^m} \sum_{T \subseteq B_q} (-1)^{a \cdot T}\, \chi_T. \tag{43}$$

The observation mask at (q, a) is a function of the m bits of one block, so its ledger has 2^m nonzero entries, and hearing $a = \psi(q)$ combines every correlator with its 2^m partners — one per toggle T within the asked block:

$$\chi'_S \;=\; \frac{\sum_{T \subseteq B_q} (-1)^{a \cdot T}\, \chi_{S \triangle T}}{2^m\, M_{a,q}}, \tag{44}$$

the denominator once more the $S = \emptyset$ instance of the numerator. For $m = 1$ the three formulas collapse to equations (37), (38) and (40).

EXAMPLE

To draw the update, take a sparse prior at $n = 3$, $m = 2$: four maps, $\phi(q) = q \bmod 4$ (weight 0.45), constant 00 (0.35), "11 iff $q \geq 4$" (0.12), constant 11 (0.08). Werner asks $q = 110$ and hears $a = 11$: toggle signs $(+, -, -, +)$ on $(\emptyset, 110_0, 110_1, \text{both})$, where q_i denotes site (q, i), and denominator $2^m M_{11,110} = 0.8$. The pairs of the $m = 1$ figure become *quartets*:

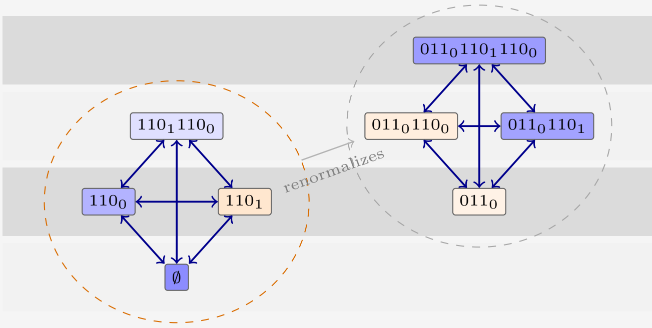


Figure 7: For $m = 2$, toggling any subset of the asked block's two sites ties four correlators into a clique that averages jointly, per equation (44). The on-block quartet (orange hull) contains the normalization and the entire block of $M_{\cdot,110}$; its signed sum is the denominator that renormalizes every other quartet. Fills to scale for the sparse prior, as before.

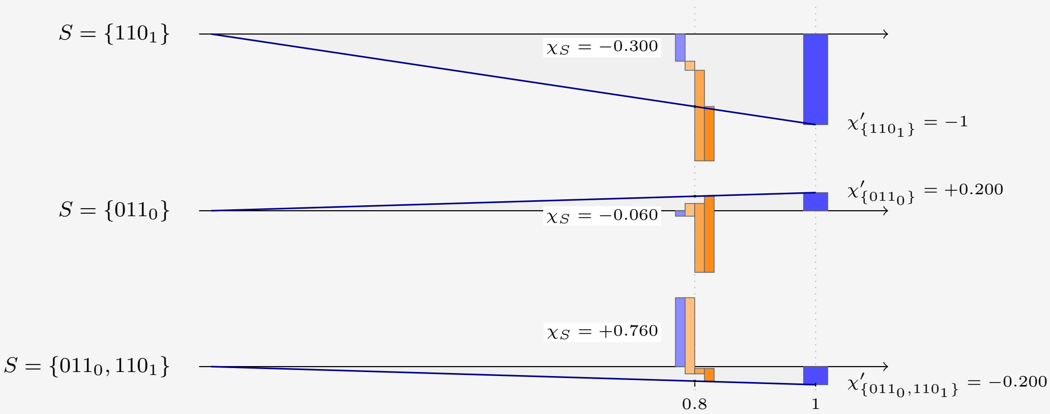


Figure 8: The $m = 2$ update, three sets from the sparse prior. The waterfall now has four segments: χ_S (blue) and its three signed quartet partners, in toggle order 110_0, 110_1, both (light to dark orange). The ray from the origin rescales the endpoint from $x = 2^m M_{a,q} = 0.8$ to $x = 1$. The asked site saturates to -1; the foreign site 011_0 flips sign, $-0.060 \to +0.200$; and $\{011_0, 110_1\}$ lands on -0.200, the slaved value $-\chi'_{\{011_0\}}$.

5 Expectations

In expectation over the truth $\psi \sim p$, as well as over the question order, we take the ratio of the expected totals,

$$\langle L_\ell \rangle \;:=\; \frac{H\big(M^{(0)}\big) - \big\langle H\big(M^{(\ell)}\big)\big\rangle}{\big\langle \sum_{i=1}^{\ell} s_i \big\rangle}\,. \tag{45}$$

We did this implicitly in Section 2.1. There, the problem was actually symmetric in $q_0 \leftrightarrow q_1$. That is not the case in general.

In order to progress, we will be interested in such averages. Both are natural. The question order is uniform, without replacement: before any evidence arrives, there is no reason to prefer one question over another. And the truth is drawn from the prior itself: Werner scores his future against his own beliefs, and he has faith in them.

The bookkeeping simplifies at once. Updating commutes, so after ℓ rounds the state depends only on the asked *set* S, a uniform random subset of size ℓ. And the received surprisals telescope to the surprisal of the whole answer block,

$$\sum_{i=1}^{\ell} s_i \;=\; -\log_2 P\big(A_S = \psi(S)\big), \tag{46}$$

whose average over ψ is a block entropy. Everything is therefore generated by one sequence, the mean block entropy at size k:

$$G_k \;:=\; \binom{2^n}{k}^{-1} \sum_{|S|=k} H(A_S), \qquad G_0 = 0, \qquad G_{2^n} = H(p). \tag{47}$$

G_1 is the mean column entropy, so $H(M)_0 = 2^n G_1$. Two exact laws follow. The first is equation (46) averaged:

$$\Big\langle \sum_{i=1}^{\ell} s_i \Big\rangle \;=\; G_\ell. \tag{48}$$

For the second, fix S and average the posterior table over the answers. Each unasked column's entropy becomes a conditional entropy, $H(A_{q'} \mid A_S) = H(A_{S\cup\{q'\}}) - H(A_S)$, and the asked columns contribute zero. Now average over S: every set of size $\ell+1$ arises as $S \cup \{q'\}$ once per element, and counting with $(\ell{+}1)\binom{2^n}{\ell+1} = (2^n{-}\ell)\binom{2^n}{\ell}$,

$$\big\langle H\big(M^{(\ell)}\big)\big\rangle \;=\; (2^n - \ell)\,\big(G_{\ell+1} - G_\ell\big). \tag{49}$$

The whole expected learning curve is the discrete derivative of one monotone sequence. By conservation, destruction and deduction are the differences $H(M)_0 - \langle H(M^{(\ell)})\rangle$ and $H(M)_0 - \langle H(M^{(\ell)})\rangle - G_\ell$, and equation (45) becomes

$$\langle L_\ell \rangle \;=\; \frac{H(M)_0 - (2^n - \ell)\big(G_{\ell+1} - G_\ell\big)}{G_\ell}\,. \tag{50}$$

In terms of the prior, the block entropies are explicit. The maps that answer the pattern b on the asked set S form a residue class, with mass

$$G_\ell \;=\; -\binom{2^n}{\ell}^{-1} \sum_{|S|=\ell} \sum_b P_{S,b} \log_2 P_{S,b}, \qquad P_{S,b} \;:=\; \sum_{j\,:\,\phi_j(q)=b_q \ \forall q\in S} p_j, \tag{51}$$

the inner sum running over the 2^ℓ answer patterns b on S. Nothing here is specific to $m = 1$: blocks of m-bit answers obey the same laws verbatim, with b running over the $(2^m)^\ell$ patterns of the block.

Some observations.

- Expectation kills the windfall. Each step has $\langle \Delta H(p) - s\rangle = 0$, so the expected deduced part is a pure tower withdrawal: $H(M)_0 - \langle H(M^{(\ell)})\rangle - G_\ell = C_0 - \langle C_\ell\rangle$, with $\langle C_\ell\rangle = \langle H(M^{(\ell)})\rangle - \big(H(p) - G_\ell\big)$. It is nonnegative: anti-leverage is only ever bad luck.
- Deduction saturates one round early: substituting $G_{2^n} = H(p)$ at $\ell = 2^n - 1$ already gives $\langle C_{2^n-1}\rangle = 0$. The store is empty before the last question, which can only be observed, never leveraged.
- $\langle L_\ell\rangle \geq 1$ always, but it need not be monotone. The endpoint is forced: $\langle L_{2^n}\rangle = H(M)_0/H(p) = 1 + C_0/H(p)$, the run ratio quoted in Section 3. This is also why equation (45) is a ratio of expectations: the expectation of the ratio is dominated by lucky runs with vanishing denominators.

- Sanity: $\ell = 0$ returns $H(M)_0 = 2^n G_1$, and at $n = 1$ the laws reproduce the one-parameter world of Section 2.1.

There is also a monotonicity theorem hiding in G. Entropy is submodular, so the increments $G_\ell - G_{\ell-1}$, which are the expected surprisals of the ℓ-th answer, decrease with ℓ: answers get cheaper as evidence accumulates. Greedy play does the opposite on purpose, seeking out the most expensive question on the board.

EXAMPLE

For the guiding prior the block-entropy sequence is $G_k = (0,\ 0.731,\ 1.436,\ 2.100,\ 2.707)$, and the laws unfold it into the expected run:

ℓ	received $\langle \sum_i s_i \rangle$	remaining $\langle H(M^{(\ell)}) \rangle$	store $\langle C_\ell \rangle$	leverage $\langle L_\ell \rangle$
1	0.731	2.113	0.137	1.110
2	1.436	1.328	0.056	1.113
3	2.100	0.608	0	1.104
4	2.707	0	0	1.081

The expected leverage humps at $\ell = 2$ and lands on $2.925/2.707 = 1.081$. The realized run of Section 3 scattered around this gentle expectation: greedy questioning and a lucky truth delivered 1.311. Figure 9 stacks the expected books.

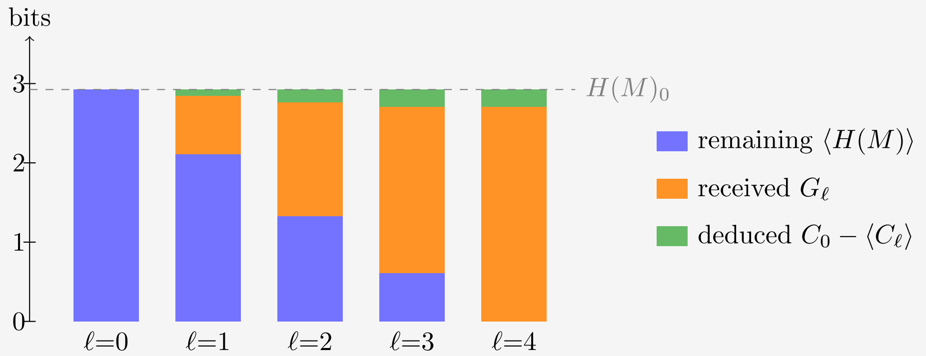


Figure 9: The expected run through the guiding example. Conservation is now exact at every ℓ: remaining plus received plus deduced reaches $H(M)_0$, so the cake always tops out on the dashed line, and no overshoot is possible. The top layer is the expected withdrawal from the correlation store, $C_0 - \langle C_\ell \rangle$; it saturates at $C_0 = 0.218$ already at $\ell = 3$.

5.1 Choosing the questions

The uniform order was a symmetry choice, not a strategy. Werner can do better along two independent axes: how far he looks ahead, and whether he commits to an order beforehand or lets each question depend on the answers so far. *Greedy* asks, at every step, the single question with the largest expected drop of $H(M)$: one step of lookahead. *ℓ-optimal* maximizes the expected knowledge after exactly ℓ questions, looking the whole horizon ahead. All four combinations are straightforward with the machinery at hand.

	order fixed beforehand	**contingent on answers**
greedy	Grow the asked set one question at a time, each maximizing the expected drop of $H(M)$ averaged over the still-unseen answers. One pass of the bookkeeping of equation (51); front-loads the cheap destruction, but cannot react to luck.	Recompute the forecasts of Section 3 after every answer and ask the current argmax: the walkthrough's play. One update per step; beats its fixed twin whenever an answer reshuffles the ranking.
ℓ-optimal	Updating commutes, so a fixed plan is just a set: search the $\binom{2^n}{\ell}$ subsets for the smallest expected remainder $\sum_{q' \notin S} H(A_{q'} \mid A_S)$. The best non-adaptive knowledge; sees correlations that the one-step view misses.	Backward induction over posteriors, $V_t = \min_q \mathbb{E}_a V_{t-1}$ with $V_0 = H(M)$. The Bayes-optimal policy: dearest to compute, and the ceiling the other three approximate.

Expected knowledge improves left to right and top to bottom. The dynamic program in the last cell is exact but grows quickly: its states are the reachable posteriors, one per pair of asked set and answer pattern, $(1 + 2^m)^{2^n}$ in all. That is 81 for the guiding example and astronomical beyond; its policy is also horizon-dependent, since the best first question for $\ell = 1$ need not open the best plan for $\ell = 2$. For the guiding prior, contingent greedy already attains the contingent optimum at every horizon: expected remaining 1.821, 0.903, 0.334 bits, against the uniform 2.113, 1.328, 0.608. That is a kindness of this prior, not a law: a prior that stores its knowledge in deep shells can defeat greedy in either column, since a question worth little now may unlock a cascade later.

6 The Gibbs complexity prior

Every prior so far was engineered: a mechanism first, a distribution second. This section builds the prior the other way around, from a single principle – simple explanations are likelier – with circuit complexity as the measure of simplicity, made quantitative as a Gibbs ensemble over the hypothesis space.

6.1 NAND complexity, and two cheaper descriptors

Fix the universal gate: the two-input NAND. The complexity X_j of a map ϕ_j is the fewest gates in any feed-forward circuit that computes it. For the bookkeeping we count in the equivalent *and-inverter* form: two-input AND gates of unit cost, with inversion free on every wire – an edge attribute, not a gate. De Morgan's laws push inverters along wires and into gates, so an and-inverter circuit converts to a NAND-only circuit of the same gate count and back; X_j is the standard AIG-size metric of logic synthesis. Reference values: AND, OR, NAND and NOR all cost 1; XOR costs 3; constants, projections and negations cost 0. For $m > 1$ outputs, X_j is the size of one *joint* circuit computing all output bits together, gates shared – the honest minimum-hardware cost, smaller in general than the sum of per-output circuits.

The tabulation is feasible because X_j is invariant under relabeling: permuting or negating inputs, negating or (for $m = 2$) swapping outputs are all free operations, so the 65,536 maps of the $(4,1)$ and $(3,2)$ spaces collapse to 222 and 308 equivalence classes. The $(4,1)$ class values are published optimal AIG sizes[1]; the $(3,2)$ values come from that synthesizer: circuit sizes $r = 0, 1, 2, \ldots$ are encoded as satisfiability instances, and the first satisfiable r is provably minimal, every smaller size having been refuted.

Alongside the interior cost X_j, which takes a SAT solver to see, two descriptors are visible in one pass over the truth table:

- the *footprint* $F_j = nI_j + S_j$, where the support S_j counts the input bits the map actually depends on and the image size I_j counts the distinct answers it attains (from 1 for constants up to 2^m). The combination is lossless – base-n digits, quotient I, remainder S – and reads as the map's *interface*: does it react to all its inputs, does it use all its values?
- the *bias* B_j: the number of 1s minus the number of 0s among the $m \cdot 2^n$ truth-table bits. This is the symmetry breaker: X_j and F_j are even under negating all outputs, B_j is odd – an energy term coupling to it acts like a magnetic field, switching on the odd correlator shells that a pure complexity prior forbids.

For the guiding $(2,1)$ space:

j	map	X_j	S_j	I_j	F_j	B_j
0	FALSE	0	0	1	2	-4
1	$\neg(b_1 \vee b_0)$ (NOR)	1	2	2	6	-2
2	$\neg b_1 \wedge b_0$	1	2	2	6	-2
3	$\neg b_1$	0	1	2	5	0
4	$b_1 \wedge \neg b_0$	1	2	2	6	-2
5	$\neg b_0$	0	1	2	5	0
6	$b_1 \oplus b_0$ (XOR)	3	2	2	6	0
7	$\neg(b_1 \wedge b_0)$ (NAND)	1	2	2	6	$+2$
8	$b_1 \wedge b_0$ (AND)	1	2	2	6	-2
9	$b_1 = b_0$ (XNOR)	3	2	2	6	0
10	b_0 (projection)	0	1	2	5	0
11	$b_1 \to b_0$	1	2	2	6	$+2$
12	b_1 (projection)	0	1	2	5	0
13	$b_0 \to b_1$	1	2	2	6	$+2$
14	$b_1 \vee b_0$ (OR)	1	2	2	6	$+2$
15	TRUE	0	0	1	2	$+4$

Grouped by the full tuple (X, F, B), the sixteen maps fall into six energy classes:

(X, F, B)	members	(X, F, B)	members
$(0, 2, -4)$	1	$(1, 6, -2)$	4
$(0, 2, +4)$	1	$(1, 6, +2)$	4
$(0, 5, 0)$	4	$(3, 6, 0)$	2

At the larger sizes the tuple breakdown runs to hundreds of rows (102 distinct tuples at $(4,1)$, 196 at $(3,2)$), so we tabulate the complexity tiers alone:

[1] The `github.com/krinkin/bounds` reproducibility dataset, cross-checked against our own SAT-based synthesizer on random classes with zero mismatches.

X	$(2,1)$	$(4,1)$	$(3,2)$
0	6	10	64
1	8	48	432
2	—	256	2,064
3	2	940	4,860
4	—	2,336	10,832
5	—	6,464	16,208
6	—	10,616	17,060
7	—	18,984	10,672
8	—	17,680	3,312
9	—	7,882	32
10	—	320	—
total	16	65,536	65,536

The tiers are thin at both ends: ten gate-free maps at $(4,1)$ (two constants, eight literals) against only 320 at the ceiling $X = 10$. The typical function is middling-complex, and it is exactly this bulk that a complexity energy pushes against. Note also that $(2,1)$ skips $X = 2$ entirely: with two inputs, nothing a two-gate circuit computes is out of reach of one gate or zero.

6.2 The ensemble

Take the energy linear in the three descriptors and the prior as its Boltzmann weight,

$$E_j = \beta\, X_j + \alpha\, F_j + \mu\, B_j, \qquad p_j = \frac{e^{-E_j}}{\sum_k e^{-E_k}}, \tag{52}$$

with no overall temperature: the couplings are the fields. Positive β suppresses complex maps, positive α suppresses large footprints, and μ couples to the signed bias, $\mu < 0$ tilting toward 1-heavy tables and thereby breaking the output-negation symmetry. Two settings are explored below:

	β	α	μ
setting A (pure Occam)	1	0	0
setting B (colder, interface-averse, tilted)	1.5	0.5	−0.2

6.3 Trajectories

Figures 10–12 show the expected cumulative leverage $\langle L_\ell \rangle$ under both settings, for the uniform-order average and the strategy grid of Section 5.1. The ℓ-optimal pair targets the halfway horizon $\ell^* = 2^n/2$ and is plotted only up to it: past its own horizon the optimizer's continuation is not defined. Inside the optimal *set*, the fixed variant asks in random order. At $(4,1)$ only the contingent optimum is dropped – its state space is $(1+2^m)^{2^n} = 3^{16} \approx 4\cdot 10^7$ – while its fixed twin, a search over $\binom{16}{8} = 12{,}870$ sets, stays affordable. Two anchors hold in every panel: all schedules agree at $\ell = 1$, because input relabeling acts transitively on the questions and the Gibbs energies are relabeling-invariant, so single questions are exchangeable; and every full-length curve ends at the same $L_{2^n} = H(M)_0/H(p)$, since any policy that asks everything receives exactly $H(p)$.

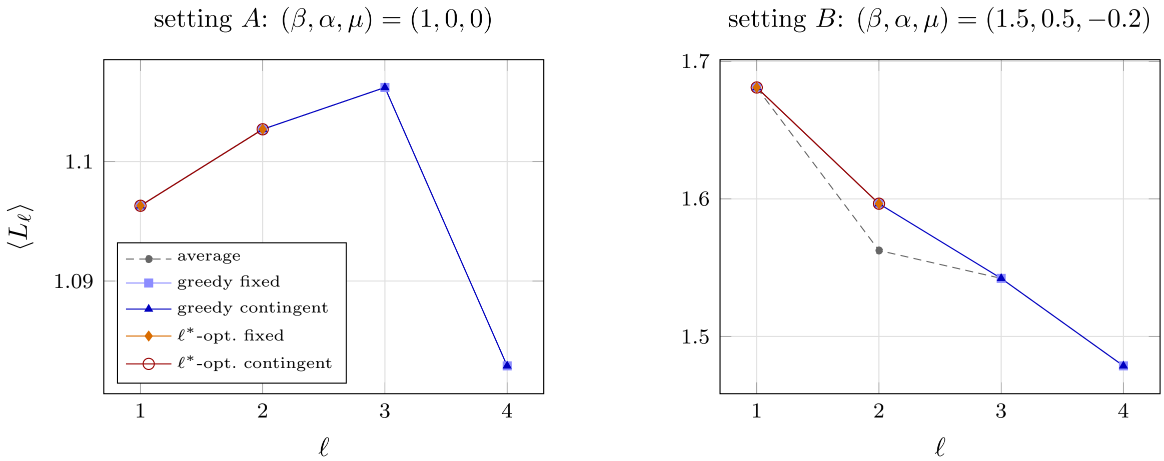


Figure 10: Expected leverage under the Gibbs prior at $(2, 1)$, $\ell^* = 2$. Left: under pure Occam all five schedules coincide exactly. Right: the fields split the schedules at $\ell = 2$.

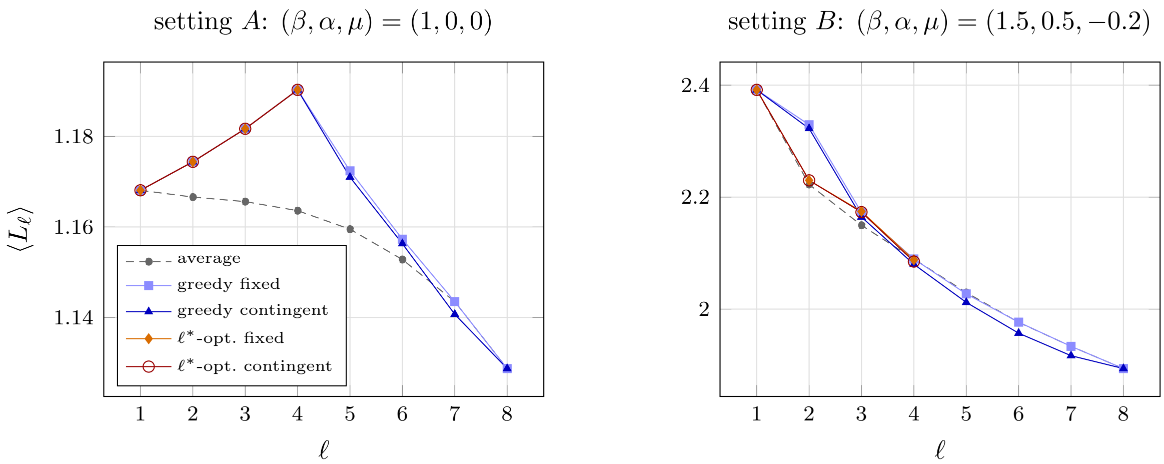


Figure 11: Expected leverage under the Gibbs prior at $(3, 2)$, $\ell^* = 4$. Right: at the horizon the ℓ^*-optimal schedules know the most and post the *lowest* leverage of the strategic four – knowledge and leverage are different objectives.

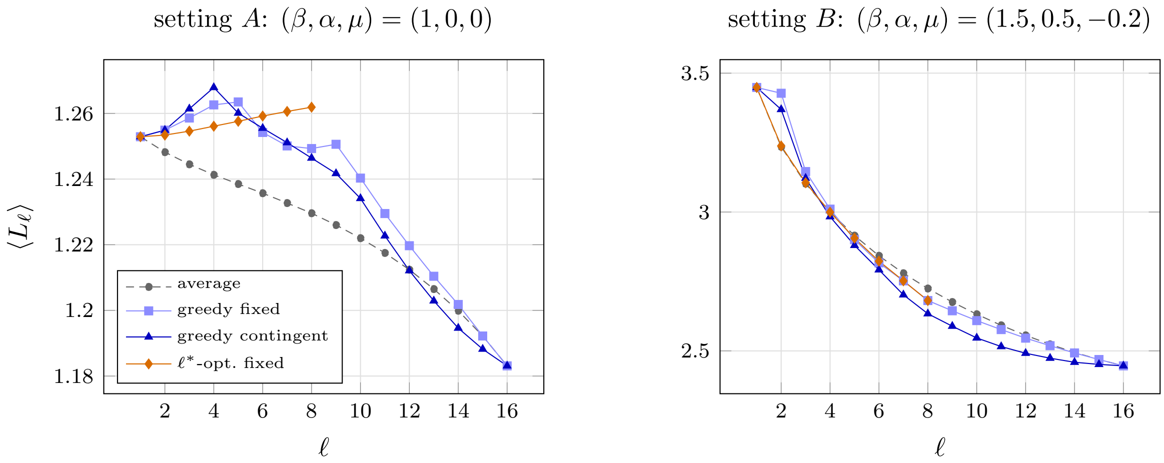


Figure 12: Expected leverage under the Gibbs prior at $(4, 1)$; only the contingent optimum is omitted. Under both settings the best fixed eight questions form a parity class of the input hypercube. The setting moves the curves by a factor of two and more; the schedule, by a few percent.

Three observations, one per system size.

At $(2, 1)$, under pure Occam, the strategy grid collapses *exactly*: all six question pairs share a single joint entropy (1.9679 bits), so every subset of every size is equivalent and every schedule is the uniform order. A complexity-only energy at this size creates correlation without preference. Setting B breaks the degeneracy – the footprint and bias fields split the pair orbits, and the strategic schedules lift above the average at $\ell = 2$ (1.5965 against 1.5625), greedy attaining the optimum. The scale jumps too: the colder ensemble stores $C_0 = 1.074$ bits against setting A's 0.306.

At $(3, 2)$, setting A rewards greed completely: the greedy fixed set is already the optimal set, and it is the four even-weight inputs $\{000, 011, 101, 110\}$ – a maximally spread design, the questions pairwise at Hamming distance two. Setting B stages the sharpest lesson of the section. The optimal set $\{001, 010, 100, 111\}$ differs from greedy's $\{000, 010, 100, 111\}$ by one question and knows 0.66 bits more at the horizon (expected remaining 1.26 against 1.92 bits) at an extra price of 0.32 received bits – and its leverage is *lower*: 2.088 against greedy's 2.090. The ℓ-optimal schedule maximizes knowledge, not leverage, and a ratio is perfectly happy with cheaper ignorance. Leverage measures how intelligently a learner spends, not how much they end up knowing.

At $(4, 1)$ the fixed optimum recovers the spread design at scale: under *both* settings the best eight questions are a parity class of the input hypercube – the even-weight inputs, pairwise Hamming distance two apart (input negation maps the class to its odd twin, an exactly equivalent design). Under setting B greedy-fixed lands on that odd twin, and the two curves merge at $\ell = 7, 8$; under pure Occam greedy's eight questions are *not* a parity class, and greedy both knows less at the horizon and posts the lower leverage (1.249 against the optimum's 1.262 – note the optimal curve is the only one that *rises* all the way to its horizon). Setting B pushes the ratio-versus-knowledge caution to its extreme: at $\ell = 8$ even the strategyless average out-levers the optimum (2.725 against 2.681) while knowing half as much (remaining 0.64 against 0.31 bits) – the typical random set is full of questions the cold prior half-knows already, so it pays 1.27 received bits where the maximally informative parity class pays 1.41. Beyond the schedules, the two settings differ far more than the schedules within either: setting A's curves sit within one percent of one another, while setting B – a cold ensemble in earnest, $H(p) = 1.68$ bits across 65,536 maps – runs from 3.45 down to

2.45. In the Gibbs world the dominant lever on leverage is the ensemble's coldness, not the question schedule.

7 The thermodynamic limit

The exact machinery lives on 2^n questions; this chapter sends $n \to \infty$. Time is measured as the asked fraction $x = \ell/2^n$, and the object of study is the limiting curve $L(x)$: what leverage survives when the question space becomes a continuum. One toy example first, then the general calculus, then a family of priors solvable in closed form.

7.1 A toy example: the spike prior

The simplest prior with something to teach: all conviction on one map, an honest hedge on the rest. Fix a special map $\phi_\star$ and a weight p, and set

$$p_\star = p, \qquad p_j = \beta := \frac{1-p}{N-1} \quad (j \neq \star), \qquad N = 2^{m2^n}. \tag{53}$$

What makes the spike worth a section is *self-similarity*: the form (53) is closed under the update rule. Asking any question leaves one of exactly two states. Either the answer confirms $\phi_\star$ – probability

$$M_\star(p) = p + \Big(\tfrac{N}{2^m} - 1\Big)\beta \;=\; 2^{-m} + \big(1 - 2^{-m}\big)\, p + O(1/N), \tag{54}$$

and the posterior is again a spike, on the $N2^{-m}$ surviving maps, with the sharper weight $p' = p/M_\star(p)$ – or it refutes $\phi_\star$, and the posterior is uniform on the survivors, an absorbing state: a uniform belief stays uniform forever. Because a single question reproduces the ansatz one size smaller, the spike lets us zoom in on the effect of *one* question and iterate. Writing q_k for the probability that the spike has survived k questions,

$$q_{k+1} = q_k\, M_\star(p_k), \qquad p_{k+1} = \frac{p_k}{M_\star(p_k)}, \qquad q_0 = 1,\; p_0 = p, \tag{55}$$

and the product is conserved, $q_k p_k = p_0$ for all k: Werner's unconditional belief in $\phi_\star$ cannot drift in expectation (the law of iterated expectations – eliminated maps carry weight zero, they do not leave the space). The invariant linearizes the recursion, and with the large-N transition (54),

$$q_k = p_0 + (1-p_0)\, 2^{-mk}, \qquad p_k = \frac{p_0}{p_0 + (1-p_0)\, 2^{-mk}} : \tag{56}$$

in odds, each confirming answer delivers exactly m bits of evidence, and $q_\infty = p_0$ – the probability of never being refuted converges to the probability of being right.

Every one-question quantity is now elementary. The two surprisals are $s_\checkmark = -\log_2 M_\star(p)$ and $s_\times = -\log_2\big(1 - M_\star(p)\big)$; the answer-matrix entropy before the question is $2^n H_{\mathrm{col}}(p)$ with the column entropy

$$H_{\mathrm{col}}(p) = -P_\star \log_2 P_\star - \big(2^m - 1\big)\, u' \log_2 u', \qquad P_\star = p + \tfrac{1-p}{2^m}, \quad u' = \tfrac{1-p}{2^m}, \tag{57}$$

and after it, $(2^n - 1)H_{\mathrm{col}}(p_1)$ on the confirming branch against $(2^n - 1)\, m$ on the refuting one. Splitting the expected drop of $H(M)$ into received and deduced, and measuring the deduced part against the store $C = H(M) - H(p)$, gives the *extraction ratio* $X_1 = D_1/C \in [0, 1]$: the proportion of the stored knowledge that one question frees.

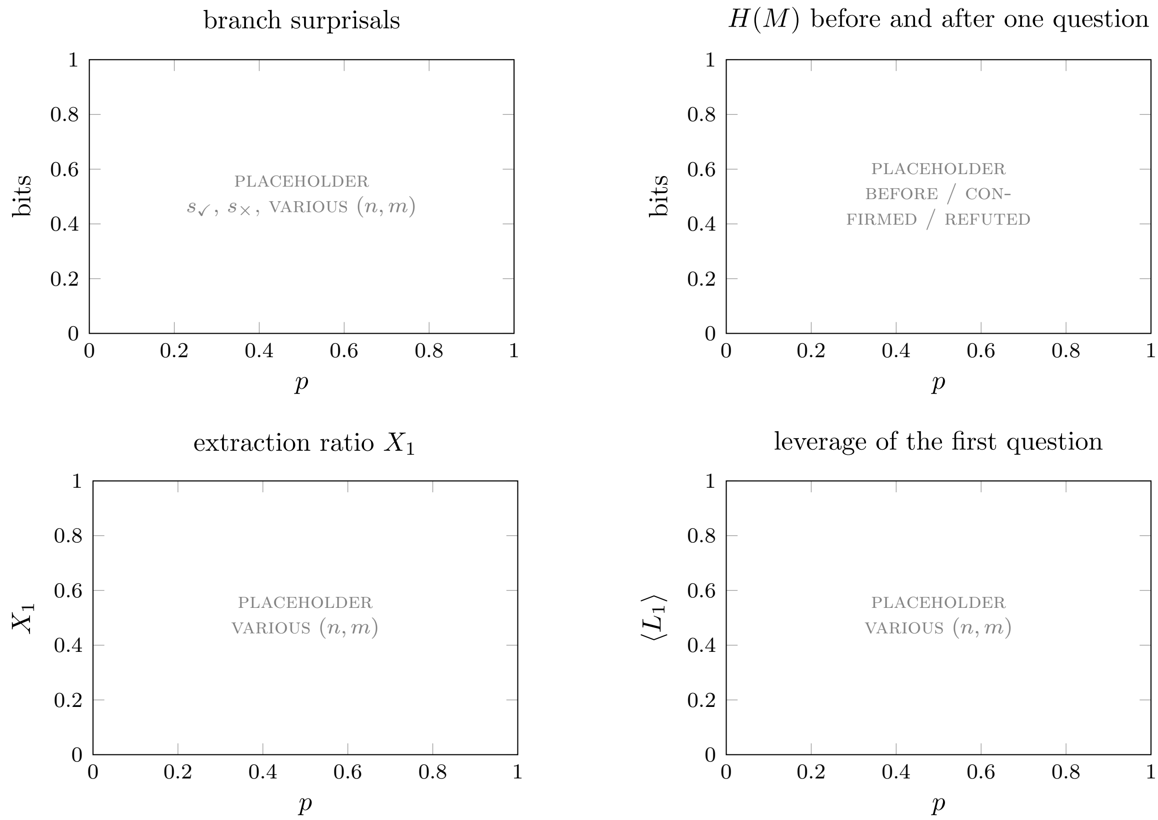


Figure 13: *Placeholder.* One-question anatomy of the spike prior as a function of the spike weight p, for several (n, m): the two branch surprisals; the answer-matrix entropy before the question and on the two branches after it; the extraction ratio; the first question's leverage.

Both ratios have clean sharp limits. As $p \to 1$,

$$X_1 \;\longrightarrow\; \frac{(2^n-1)\big(1-2^{-m}\big)^2}{2^n\big(1-2^{-m}\big)-1} \xrightarrow{n\to\infty} 1-2^{-m}, \qquad \langle L_1\rangle \;\longrightarrow\; 2^n-(2^n-1)\,2^{-m}, \tag{58}$$

so the first question's leverage is *extensive*, and scaled by the question count it converges to the same constant:

$$\frac{\langle L_1\rangle}{2^n} \xrightarrow{p\to1,\; n\to\infty} 1-2^{-m}. \tag{59}$$

The constant is the *culling ratio*: one answer kills the fraction $1-2^{-m}$ of hypothesis space, and for a near-delta prior that is also the fraction of every column the answer settles. The deficit 2^{-m} is the probability that total ignorance fakes the expected answer. The limit is not decoration but necessity: at $p = 1$ exactly the prior has $H(p) = 0$, the first question receives nothing and frees nothing, and L_1 is the undefined $0/0$. The spike weight must approach certainty without reaching it, and the approach is logarithmically slow – the price of a prior whose entire store sits one answer deep.

The other order of limits gives the whole trajectory. Fix $p < 1$ and send $n \to \infty$ at asked fraction $x = k/2^n$: the spike branch sharpens to a delta within $O(1)$ questions – a vanishing fraction – so at every $x > 0$ the two branches have long settled, and the cumulative leverage converges to a pure hyperbola,

$$\big\langle L_{x2^n}\big\rangle \xrightarrow{n\to\infty} 1+\frac{c(p,m)}{x}, \qquad c(p,m) = \frac{H_{\mathrm{col}}(p)-(1-p)\,m}{(1-p)\,m}, \tag{60}$$

with H_{col} from (57). The reading: the entire store is freed inside a vanishing fraction of the run, and what the curve records afterwards is that early windfall diluting into the growing received total. The two orderings do not commute – $p \to 1$ first gives the extensive plateau (59), flat in x; $n \to \infty$ first gives a finite hyperbola for every $p < 1$, with c diverging logarithmically as $p \to 1$.

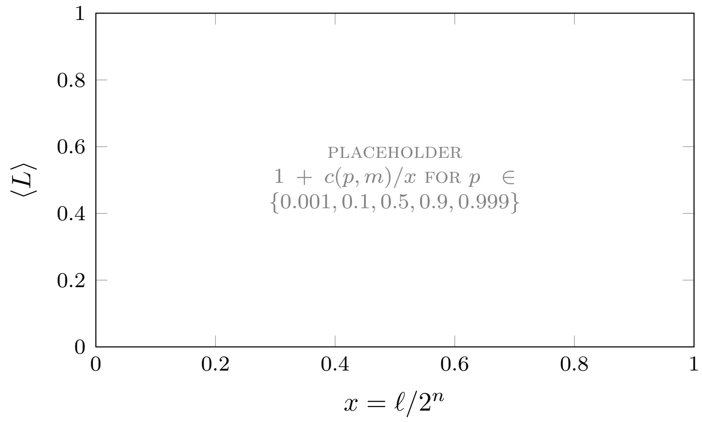


Figure 14: *Placeholder.* The limiting spike leverage (60) against the asked fraction, for spike weights spanning six orders of hedging. All curves share the $1/x$ shape; the weight only sets the coefficient.

7.2 Thermodynamic limit calculus

The spike's hyperbola is one instance of a general calculus. Recall the two exact laws of Section 5: under a random question order and a truth drawn from the prior, the expected received after ℓ questions is the mean block entropy G_ℓ, and the expected remaining answer-matrix entropy is $(2^n - \ell)(G_{\ell+1} - G_\ell)$. Everything in the limit is generated by the increments. Define the *profile*

$$\gamma(x) := \lim_{n\to\infty} \left(G_{\lfloor x2^n \rfloor+1} - G_{\lfloor x2^n \rfloor}\right), \tag{61}$$

the expected entropy of one more answer given a random x-fraction of the table. By the chain rule this is a conditional entropy, so it is nonincreasing in x and trapped in $[0, m]$: monotone and bounded, the best-behaved objects in analysis. Telescoping, $G_\ell = \sum_{k<\ell} \gamma_n(k)$ is a Riemann sum of the staircase $t \mapsto \gamma_n(\lfloor t2^n \rfloor)$, and if the staircase converges pointwise the bounded-convergence theorem passes it to the integral. Dividing the laws by 2^n:

$$\frac{\text{received}}{2^n} \to g(x) := \int_0^x \gamma(t)\,dt, \qquad \frac{\text{remaining}}{2^n} \to (1-x)\,\gamma(x), \qquad L(x) = \frac{\gamma(0) - (1-x)\,\gamma(x)}{g(x)}. \tag{62}$$

Endpoints by first-order expansion: $L(0^+) = 1 - \gamma'(0)/\gamma(0) \geq 1$ for a differentiable profile, and $L(1) = \gamma(0)/g(1) = H(M)_0/H(p)$, the whole-run ratio.

Three assumptions were spent, and each names a way for a prior family to break the limit. *Extensivity*, at the division: $H(p^{(n)})$ must scale like 2^n, or the learning lives on a sublinear clock and L diverges. *Self-averaging*, at the pointwise convergence: the staircases of the family $p^{(n)}$ must settle on a single profile rather than oscillate with n. *Positivity*, $\gamma > 0$ on $[0, 1)$, not needed for the limit but needed for $L(x)$ to stay regular: a profile that hits zero at $x^* < 1$ compresses an extensive avalanche of deduction into that instant. Nothing in (61)–(62) chooses a prior: γ is computed from any concrete $p^{(n)}$ through its block entropies, and the rest of this chapter is a gallery of priors whose γ comes out in closed form.

7.3 The coin-flip prior

The simplest scaling family, worked like a calculus exercise. Take R coins with biases θ_r and mixing weights w_r: nature picks coin r with probability w_r, then answers every question independently, each answer bit 1 with probability θ_r. A map with W_j ones among its $m2^n$ table bits gets

$$p_j = \sum_{r=1}^{R} w_r \, \theta_r^{W_j} (1-\theta_r)^{m2^n - W_j}. \tag{63}$$

At $(2,1)$ with the running example $\theta \in \{0.2, 0.7\}$, $w = (\frac{1}{4}, \frac{3}{4})$, the sixteen maps get weights ranging from 0.0395 (the half-and-half tables, which neither coin likes) to 0.1805 (TRUE): [worked table to be placed here alongside the derivation steps].

The derivation runs in four steps, each elementary. *One*: at $\ell = 0$ the predictive is the mean coin, $\gamma(0) = h(\bar\theta)$ with $\bar\theta = \sum_r w_r \theta_r$. *Two*: after ℓ answers with k ones, the posterior odds between two coins contain the factor $\ell\big[\frac{k}{\ell}\log\frac{\theta_s}{\theta_r} + (1-\frac{k}{\ell})\log\frac{1-\theta_s}{1-\theta_r}\big]$; by the law of large numbers $k/\ell \to \theta_{\text{true}}$, the bracket converges to a negative relative entropy, and the wrong coins die exponentially. *Three*: the predictive converges to the true coin, so for every fixed $x > 0$ the profile equals the mean coin entropy, $\gamma(x) = \bar g := \sum_r w_r h(\theta_r)$ – a step function, $h(\bar\theta)$ at the single point $x = 0$ and $\bar g$ after. *Four*: a single point has measure zero, so $g(x) = \bar g x$, and the master formula collapses to

$$L(x) = 1 + \frac{c}{x}, \qquad c = \frac{h(\bar\theta) - \bar g}{\bar g} \;\geq\; 0 \tag{64}$$

by Jensen's inequality on the concave h. One coin gives $c = 0$, the product prior, $L \equiv 1$: trivial. And two coins already exhaust the family: the limit sees the mixture only through the pair $(h(\bar\theta), \bar g)$, two atoms sweep every admissible pair by the intermediate value theorem, and de Finetti's theorem closes the class – any consistently exchangeable family *is* a coin mixture. In the thermodynamic limit, two parameters is all the richness exchangeability has.

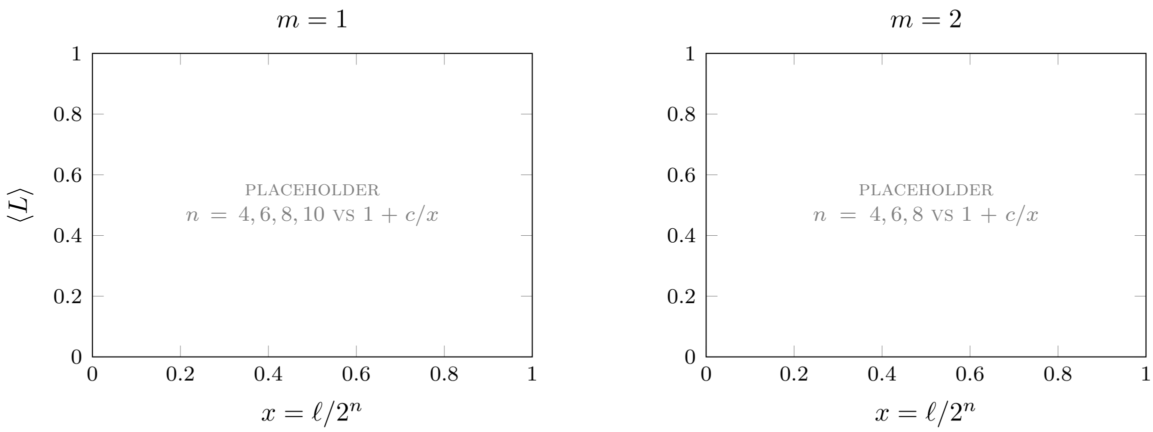


Figure 15: *Placeholder.* Exact finite-n coin-mixture leverage converging onto the hyperbola (64), at two answer widths.

The family line is worth drawing: the coin mixture and the spike are the same mechanism. Both hide *one global secret* – which coin, which map – read off within a vanishing fraction of the run; both then coast on independent leftovers. That is why both land on $1 + c/x$: the hyperbola is not a shape the prior chooses but the signature of front-loaded deduction diluting into the denominator. Escaping it requires a prior whose secrets cannot all be read at the start.

7.4 Non-hyperbolic solvable priors

What must break is *exchangeability*. A prior invariant under permuting the questions is, consistently across n, a coin mixture, and Section 7.3 showed those only produce hyperbolas. Non-hyperbolic limits therefore need priors that treat blocks of questions differently – correlations with an address.

The construction: partition the 2^n questions into disjoint *blocks* of size r; draw each block's answers independently, from a distribution symmetric under permuting the block's members. Independence makes block entropies add; within-block symmetry makes them depend only on how many of the block's questions have been asked. Writing $\eta(i)$ for the entropy of one more member given i already-asked members of its block, the profile follows from a single limit – the number of asked partners of a fresh question is hypergeometric, and sampling without replacement from an enormous pool becomes binomial:

$$\gamma(x) = \sum_{i=0}^{r-1} b_i(x)\,\eta(i), \qquad b_i(x) := \binom{r-1}{i} x^i (1-x)^{r-1-i}. \tag{65}$$

The b_i are the *Bernstein basis polynomials*: degree $r-1$, positive on $(0,1)$, summing to one, the i-th peaked at $x = i/(r-1)$. At degree three, for instance,

$$b_0 = (1-x)^3, \quad b_1 = 3x(1-x)^2, \quad b_2 = 3x^2(1-x), \quad b_3 = x^3. \tag{66}$$

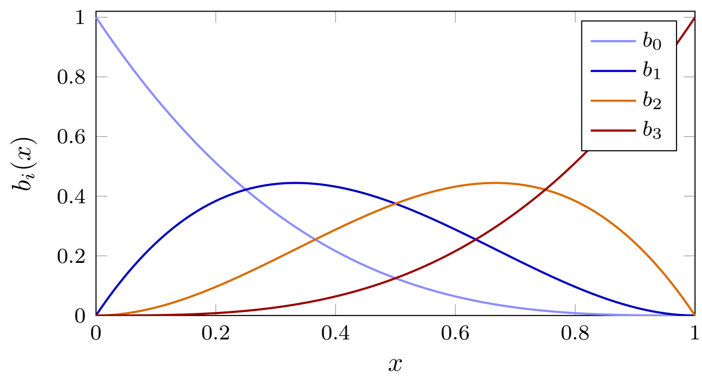


Figure 16: The degree-three Bernstein basis (66): bump functions scanning left to right, each peaked at $i/3$. A block profile is a positive combination of such bumps, weighted by the microscopic conditional entropies $\eta(i)$.

Equation (65) is the whole theory of block priors: the profile is a Bernstein polynomial whose coefficients are the block's microscopic conditional entropies, necessarily nonincreasing in i. The shape of L is then a question about *where the coefficients drop*. Coefficients falling at $i = 0$ front-load the deduction (hyperbola-like decay); a drop at the last index back-loads it and $L(x)$ *rises*; a drop at an interior index concentrates the windfall mid-run, and with fresh entropy after it the leverage becomes non-monotonic – a genuine interior peak, impossible for any exchangeable prior.

7.5 An exotic zoo of solvable priors

Five members, each solvable because its $\eta(i)$ is explicit. For each: the block rule, the profile, the shape.

Cliques. All r members share one value, drawn from an arbitrary block-specific distribution ν: $\eta = (H(\nu), 0, \ldots, 0)$, so $\gamma(x) = \langle H(\nu)\rangle (1-x)^{r-1}$ and the scale cancels between destroyed and received: $L \equiv r$, exactly flat – every question is either the first touch of its block (one value received, r columns collapse) or a free repeat.

Parity blocks. Uniform on the strings of fixed parity β: any partial view is uniform, so $\eta = (1, \ldots, 1, 0)$, $\gamma(x) = 1 - x^{r-1}$, and L rises monotonically from 1 to $r/(r-1)$: the windfall fires only at a block's last member, deduction back-loaded.

Tilted parity. Draw the block's bits i.i.d. Bernoulli(θ) conditioned on the parity: the general form, with $\varepsilon := 1 - 2\theta$ as the new parameter. The conditionals stay closed-form; $\eta(i) = h(\theta) + O(\varepsilon^{r-i})$ far from the cliff and $\eta(r-1) = 0$ exactly: the uniform case is $\varepsilon = 0$, and the tilt lets partial views leak $O(\varepsilon^{r-i})$ bits early, softening the onset without touching the final deduction.

MDS blocks. The block carries a uniformly random codeword of a code where any k symbols determine all r and any $j \leq k$ reveal nothing (Reed–Solomon over 2^m-ary answers): $\eta = m$ up to $i = k-1$, then 0, so $\gamma(x)/m = \Pr[\mathrm{Bin}(r-1, x) \leq k-1]$: a movable threshold at $x \approx k/r$. Diluted with fresh questions the curve climbs to an interior peak and relaxes – the non-monotonic shape promised above.

The cocktail. A quarter coins, half MDS, a quarter fresh, on disjoint blocks: profiles average, and the curve inherits all three mechanisms in sequence – a hyperbolic dive, a dip, a resurgent peak, a final decay.

None of the hard members has full support, and all can be given it without disturbing the limit. The recipe is *local* noise: flip bits with probability δ inside a block, or lapse each block to an i.i.d. one with probability δ – never a latent shared across blocks. Locality preserves independence and within-block symmetry, so the machinery applies verbatim and the noise acts term by term on the coefficients: the clique's $\eta(i \geq 1)$ lifts from 0 to the $O(h(2\delta))$ scale; the parity cliff $\eta(r-1)$ lifts to $h\big(\frac{1-(1-2\delta)^r}{2}\big)$; the MDS tail lifts to $\approx \delta m$. At fixed δ the limit exists and is a continuous deformation of the hard one – the limit depends on the noise only through the finitely many $\eta(i)$ – and under any vanishing schedule $\delta_n \to 0$ the prior has full support at every finite n while the limit is *exactly* the hard curve. What must be avoided is global softening: a shared latent with extensive entropy floods the received channel and can erase the store outright.

Nothing here is finely tuned. An *arbitrary* mixture of block types – fractions w_t of questions in blocks of size r_t with any within-block-symmetric distributions – is solvable by the same one formula,

$$\gamma(x) = \sum_t w_t \sum_{i=0}^{r_t-1} b_i^{(r_t-1)}(x)\, \eta_t(i), \tag{67}$$

symbolically, before any constituent is specified: the $\eta_t(i)$ are the only inputs, and mixing blocks mixes their profiles linearly (the leverage, a ratio, then mixes as a mediant). The solvability is a property of the block structure, not of a lucky choice of members.

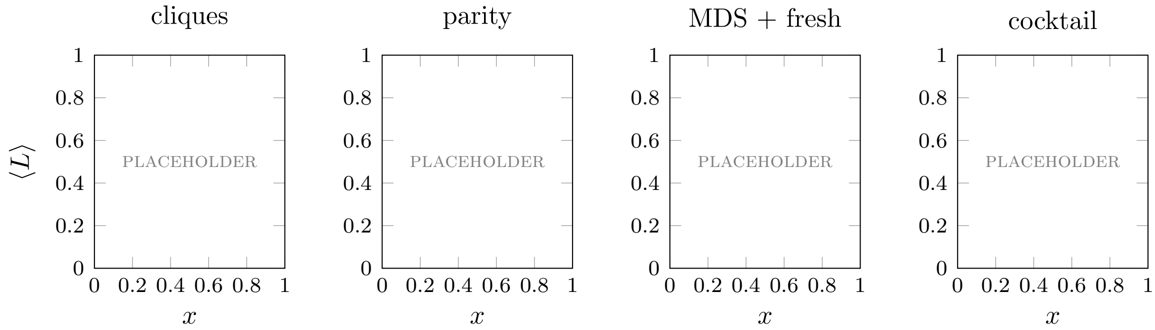


Figure 17: *Placeholder.* The zoo side by side: flat, rising, peaked, and the cocktail's dive–dip–peak–decay, each with exact finite-n curves converging on its limit.

member	block rule	$\eta(i)$	$\gamma(x)$	$L(x)$
cliques	one shared value ν	$H(\nu), 0, \ldots$	$\langle H(\nu)\rangle(1-x)^{r-1}$	$\equiv r$, flat
parity	uniform, $\oplus = \beta$	$1, \ldots, 1, 0$	$1 - x^{r-1}$	rises to $\frac{r}{r-1}$
tilted parity	i.i.d. θ given $\oplus = \beta$	plateau, cliff	closed form in ε	rises, soft onset
MDS + fresh	uniform codebook	$m, \ldots, m, 0, \ldots$	$w\bar{F} + (1-w)$	peak near k/r
cocktail	coins, code, fresh	mixture	mixture	dive, dip, peak

The zoo earns a closing remark. Each member's limiting curve is the visible fingerprint of an invisible structure: hyperbolas betray one global secret, flat lines betray duplication, rising curves betray constraints that bind only when complete, peaks betray thresholds. The inference runs backwards as well as forwards. A well pre-trained model traces an empirical $L(x)$ as it learns, and the shape of that learning curve – falling, flat, rising, peaked – is evidence about which class of prior the real world's regularities are drawn from. Measuring how a learner's leverage evolves may be the most practical window we have onto the distribution type of the world itself.